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Relational Alpha Factors with Graph Attention Networks

A Dual-Track Study on US Equities and
European Power Markets

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基于图注意力网络的关系型Alpha因子

美国股票市场与欧洲电力市场双轨制研究

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Abstract

Cross-sectional alpha factors score assets in isolation, yet the assets are coupled — statistically in equity markets, physically in European power markets. This report measures what that coupling structure is worth. To make the measurement interpretable, the information set is held fixed and two questions are separated: what a graph adds, measured against an unlearned uniform-propagation anchor, and what learned attention adds, measured as a graph attention network’s margin over that anchor on the same topology. One GAT kernel, one leakage-controlled evaluation harness, and one suite of research gates run on two structurally different graphs: an estimated correlation backbone over 49 US equities and the physical interconnector network over 20 European bidding zones.

On equities, learned attention beats unlearned propagation in 30 of 30 seeded runs; the best configuration reaches an out-of-sample rank-IC of 0.0148 ± 0.0043 and long-short Sharpe of 1.37 ± 0.39 over five seeds. The resulting composite factor is unique, consistent, and robust — three of four gates — but never beats the best single input factor. Attention analysis explains the modesty: the model applies a stationary, near-uniform reweighting of neighbour pooling, not sharp selection. On energy, the alpha framing failed in an instructive way: three nested evaluation artifacts — overlapping labels, wrong annualisation, and a clip on the evaluation return that concealed short-leg tail losses — each produced spectacular numbers before being caught by the project’s own controls, and under honest returns the strategy loses money. Reframed as forecasting, the energy track yields the cleanest relational evidence: the interconnector graph lifts node-level price forecast skill by +0.131 over an identically informed no-graph baseline, and on cross-border spreads — a target that does not exist without the relation — message passing beats a both-endpoint baseline in five of five seeds (+0.056 skill).

The report’s contributions are the isolation methodology, the two measured results, a worked three-artifact cautionary example for GNN-based market research, and the platform engineering that makes all of it reproducible from archived artifacts.

摘要

横截面 α 因子法单独对资产进行评估，但实际上这些资产之间存在耦合关系——在股票市场中这种耦合是统计意义上的，在欧洲电力市场中则是物理意义上的。本报告旨在衡量这种耦合结构的价值。为使该度量结果具有可解释性，本报告保持信息集不变，并将问题拆分为两个部分：一是图结构所贡献的值（以未学习的均匀传播锚点为基准进行衡量）；二是通过图注意力网络在相同拓扑结构上相对于该锚点所获得的边际增益（即图注意力网络带来的增量）。本研究采用一个GAT核函数、一个带泄漏控制的评估框架以及一套研究用门控机制，分别在两个结构不同的图上进行分析：一个是涵盖49只美国股票的估计相关性骨干网络，另一个是涵盖20个欧洲投标区的物理互联网络。

在股票投资领域，经过学习的注意力机制在 30 次随机种子样本的跑产中均表现优于未经学习的传播机制；最佳配置在五组随机种子样本上达到了样本外 Rank-IC 值为 0.0148 ± 0.0043 以及长短期 Sharpe 指数为 1.37 ± 0.39 。由此得出的复合因子具有独特性、一致性和稳健性——在四项门控因子中三项表现优异——但始终未能超越最佳单输入因子。这种表现平平的原因可通过注意力机制分析来解释：该模型对邻域聚合采用了平稳的、近乎均匀的重新加权方法，而非尖锐的选择机制。在能源领域，“Alpha 框架”策略的表现揭示了一个具有启发意义的现象：三种嵌套的评估偏差——重叠标签、错误的年度化处理以及对评估收益率施加的截断处理（该处理掩盖了短端尾部损失）——在各自单独出现时均产生了极高的数值，但最终均被该项目自身的控制变量所遏制；而在真实收益率环境下，该策略实际出现了亏损。当将该策略重新定义为预测模型时，其“能源赛道”策略提供了最清晰的关联性证据：互联网络图将节点层面的价格预测能力提升了 +0。在与之信息对称的无图基线模型上，以及在跨境利差预测任务中——这一目标若无该关联则无法实现——消息传递方法在全部五组种子数据中均优于双端点基线模型（技能提升 +0.056）。

本报告的贡献包括：数据隔离方法论、两项实测结果、一个基于图神经网络（GNN）开展市场调研的、包含三个人工构建样本的示例性案例，以及一套可使整个研究过程基于存档的构建样本实现可复现性的平台工程方案。

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1 Introduction

Quantitative alpha research scores assets one at a time. A formulaic alpha reads a single stock’s prices and volumes and emits a number; the cross-section of those numbers, ranked, is the signal [1]. Yet the assets themselves are anything but isolated: equities co-move within sectors and supply chains, and European power prices are literally wired together by cross-border transmission lines. The premise of this project is that the coupling structure is information, and its subject is a careful measurement of how much.

The measurement question splits in two, and the split matters. That a graph-augmented model outperforms a non-relational baseline is easy to show and hard to interpret — the gain may come from the graph, from the extra parameters, or from information the baseline never saw. This project therefore fixes the information set (the same island alphas every model consumes) and separates two claims: what the *graph* adds, measured by an unlearned uniform-propagation anchor, and what *learned attention* adds, measured as the gap between a graph attention network [2] and that anchor on the same topology. Three research questions organise the work:

- RQ1 Does learned attention add value over unlearned relational propagation, holding inputs and topology fixed?
- RQ2 Does one GAT kernel and one evaluation harness transfer across structurally different graphs — an estimated correlation backbone over US equities and a physical interconnector network over European bidding zones — without manufacturing false positives?
- RQ3 On real power-market data, does the physical topology improve forecast skill, and where does its value concentrate?

A reader from outside quantitative finance should carry one picture through what follows (Section 2.1 defines the recurring vocabulary). Every model in this report is trained on the same task: score today’s cross-section of assets so that the scores rank the next period’s price moves, five trading days ahead for equities, one day ahead for power. Every model in a comparison also receives the same per-asset inputs: ten signals computed from each stock’s own prices and volumes on the equity track, and each zone’s price dynamics plus its published fundamentals (load, wind and solar forecasts among them) on the energy track. In particular, the graph neural networks here are not a vehicle for feeding extra data into price prediction — whatever exogenous drivers exist go to the graph-free baselines too. The relational models’ only privilege is topological: permission to exchange information along edges, estimated return correlations for equities and the physical cross-border grid for power. Every headline number in this report is therefore a margin, the value of that permission with everything else held fixed.

The answers, in brief. On the equity track, learned attention beats the uniform anchor in thirty of thirty seeded runs, and the best configuration reaches an out-of-sample (OOS) IC of 0.0148 ± 0.0043 and Sharpe of 1.37 ± 0.39 over five seeds; the resulting composite passes three of four research gates but never beats the single best island alpha, a failure this report states as plainly as the successes (Chapter 5). On the energy track the alpha framing collapsed honestly: three nested evaluation artifacts each produced spectacular numbers before the project’s own controls caught them, and under honest returns the strategy loses money (Chapter 6). Reframed as forecasting, the same track produced the project’s cleanest positives: the interconnector graph lifts node-level price forecast skill by +0.131 over a no-graph baseline with identical drivers, and on the irreducibly relational target — cross-border spreads — message passing beats a both-endpoint baseline in five of five seeds (Section 6.6).

1 引言

定量阿尔法研究方法对各类资产进行逐一评估。一种基于公式化的阿尔法方法会分析单只股票的价格与成交量，并输出一个数值；而对这些数值进行排序所得到的集合即为信号[1]。然而，各类资产本身绝非孤立存在：股票在不同行业和供应链中同步波动，而欧洲的电力价格实际上通过跨境输电线路紧密相连。本项目的核心假设是：这种耦合结构本身即蕴含着信息，而其研究对象则是对这种信息含量进行细致的量化测度。

该测量问题被拆分为两个部分，而这种拆分至关重要。一个带有图结构增强的模型性能优于非关系型基线模型这一点很容易证明，但其背后的原因却难以解释——这种性能提升可能源于图结构本身、额外的参数，或是基线模型从未接触过的信息。因此，本项目固定了信息集（即所有模型均使用的相同岛屿阿尔法值），并将两个论点分开讨论：一是图结构所贡献的部分（通过一个未经过学习的均匀传播锚点进行衡量），二是学习到的注意力机制所贡献的部分（通过在相同拓扑结构上比较图注意力网络[2]与该锚点之间的差异来衡量）。本研究围绕三个科学问题展开：

RQ1：在输入和拓扑结构保持不变的情况下，习得性注意力是否比非习得性关系传播更具价值？

RQ2：单个 GAT 内核与单个评估框架能否跨结构不同的图进行迁移？

——一个覆盖美国股票市场的估计相关性骨干网络，以及一个覆盖欧洲投标区的物理互联网络——是否能避免产生误报？

RQ3：在实际电力市场数据上，物理拓扑结构能否提升预测精度？其价值主要集中于哪些方面？

非量化金融领域读者在阅读下文时，建议参考一张示意图（第2.1节定义了相关专业术语）；本报告中的每个模型均基于同一任务进行训练：对当前资产组合进行评分，从而对下一交易周期内的价格走势进行排序——对于股票市场，该预测覆盖五个交易日；对于电力市场，预测覆盖一个交易日。在对比分析中，每个模型均接收相同的资产级输入数据：在股票市场中，包含根据每只股票自身价格和成交量计算出的十项信号；在电力市场中，包含各区域的价格动态及其公开的基本面数据（包括负荷、风电和太阳能预测等）。特别需要指出的是，本文所采用的图神经网络并非用于向价格预测模型输入额外数据——无论是否存在外生驱动因素，这些因素同样适用于无图基线模型。关系型模型所独有的优势在于其拓扑结构：即允许沿边边交换信息、为股票市场估算回报率相关性、以及为电力市场构建物理跨境电网模型。因此，本报告中的每一个关键数值都代表一个边际效应——即在其他条件保持不变的情况下，该信息交换权限所对应的数值。

简要回答：在权益投资赛道上，学习型注意力策略在全部30次种子数据集测试中均优于均匀锚定策略；最佳配置在五次种子数据集上实现了 0.0148 ± 0.0043 的样本外IC值和 1.37 ± 0.39 的夏普比率；该综合策略通过了四项研究筛选中的三项，但始终未能超越单个最佳岛屿阿尔法策略——这一失败与本报告中所记载的成功案例同样清晰（第5章）。在能源投资赛道上，阿尔法构建方法存在明显缺陷：三次嵌套的评估伪象在项目自身的控制变量捕获之前均产生了惊人的数值，而在真实回报环境下，该策略实际出现亏损（第6章）。当将该策略重新定义为预测模型时，该赛道呈现出最清晰的积极结果：互联网图谱使节点级价格预测能力较采用相同驱动因素的无图谱基线提升了+0.131；而在不可约关系型目标——跨境利差——上，消息传递策略在全部5次种子数据集测试中均优于双端点基线（第6.6节）。

Five contributions follow from this, each defended in a named part of the report. First, an isolation methodology for relational factors: matched no-learning anchors, four research gates, and structural leakage controls, defined once in Chapter 4 and reused unchanged by every experiment. Second, the equity result itself — attention’s margin over naive propagation as a distribution over seeds, devices, and configurations rather than a point estimate, together with an interpretability analysis showing the mechanism to be a gentle, stationary reweighting of mean pooling (Chapter 5). Third, a worked cautionary example: the energy alpha arc of Chapter 6, in which the project’s own controls catch three artifacts in sequence, ending in a verdict — rank correlation can be positive while the strategy loses money — that generalises beyond power markets. Fourth, the forecasting reframe and its control-validated graph value, concentrated on the relational target (Sections 6.5 and 6.6). Fifth, the platform engineering that makes the rest reproducible: two seams that let island and relational factors share one pipeline, warehouse marts built from research runs, and a test suite that pins the leakage-critical logic (Chapter 3).

A note on standards. This is a Master’s capstone report, and the working assumption throughout is that its most valuable output is not a headline Sharpe but a defensible measurement process. Wherever a result died under scrutiny, the scrutiny and the death are in the text; the negative results are load-bearing, not confessional.

The report proceeds as follows. Chapter 2 places the work in the alpha-research, GNN, and electricity-forecasting literatures. Chapter 3 describes the platform and the two seams. Chapter 4 fixes the measurement apparatus. Chapters 5 and 6 report the two tracks. Chapter 7 confronts threats to validity, and Chapter 8 closes with what a successor project should do differently.

2 Background and Related Work

2.1 Alpha research in plain terms

This report uses a small vocabulary that is standard in quantitative finance and opaque outside it; this section defines it once, so that the rest of the report does not presuppose a finance background.

An *alpha factor* is a rule that assigns every asset a score at every point in time, constructed so that higher-scored assets should subsequently outperform lower-scored ones. The prediction is about *relative* performance within a fixed universe, not about price levels: the object of interest is the *cross-section* — all assets observed at one timestamp — and whether today’s ranking of scores anticipates the ranking of returns realised over the following days. A factor that scores an asset from that asset’s own history alone (five-day price reversal, abnormal trading volume) is an *island factor* in this report’s terminology; a *relational factor* may also read the asset’s neighbours in a graph.

Two measurements recur on every page of Chapters 5 and 6. The *information coefficient* (IC) is the correlation between the scores issued at a timestamp and the returns realised afterwards; this report uses the rank form (*rank-IC*, a Spearman correlation), computed cross-sectionally at each timestamp and averaged. A rank-IC of 1 would be a perfect ordering; 0 is no predictive skill at all. Real factors are weak — the best out-of-sample composite in this report reaches a rank-IC of about 0.015 (Chapter 5) — and factors of this strength are useful only in combination, which is why the field builds them in large numbers. The *Sharpe ratio* evaluates a factor as a trading strategy instead: buy the top-ranked assets, sell short the bottom-ranked ones (a *long-short* portfolio, whose exposure to the overall market direction largely cancels), and divide the strategy’s mean return by its volatility, annualised. The two numbers can disagree, because ranking well says nothing about the magnitudes of wins and losses; Chapter 6 turns exactly this disagreement into one of the report’s main findings.

“Alpha” itself, in the industry idiom this report inherits from Kakushadze’s catalogue of practitioner formulas [1], names nothing deeper than a signal expected to carry predictive information beyond passive market exposure. No claim below depends on a stricter definition.

由此可得出五项研究成果，每项均在报告中相应的章节中进行了论证。首先，一种用于分析关系性因素的隔离方法论：包括匹配的无学习锚点、四个研究门控以及结构泄漏控制措施——这些内容在第4章中定义一次，随后在所有实验中均保持原样重复使用。其次，公平性结论本身——即注意力机制相对于朴素传播方式的边际优势，被视作种子、设备及配置的分布而非单点估计；此外还包含一项可解释性分析，揭示该机制实为对均值池化进行的一种温和、平稳的重新加权操作（第5章）。第三，一个具有警示意义的实证案例：第6章中的“能源Alpha弧”案例——在该案例中，项目自身的控制变量依次捕捉了三种人为偏差，最终得出一个结论：即使等级相关系数为正，该策略仍可能产生亏损——这一结论可推广至电力市场之外的领域。第四，预测框架及其经控制变量验证的图论价值，该部分聚焦于关系性目标（第6.5节与第6.6节）。第五，实现其余部分可复现性的平台工程：两条接缝使得岛状与关系型因素能够共享同一数据流水线；基于研究性运行构建的仓库集市；以及一个用于锁定泄漏敏感逻辑的测试套件（第3章）。

关于标准的说明：本报告为一篇硕士毕业设计论文；其贯穿始终的基本假设是：本报告最具价值的产出并非某个亮眼的夏普比率，而是一个可辩护的计量分析流程。凡结果经严格检验后未能通过之处，均在正文中予以说明；这些否定性结果具有支撑作用，而非单纯的自我交代。

本报告结构如下：第2章将本研究置于Alpha Research、图神经网络（GNN）及电力负荷预测的相关文献背景中；第3章介绍研究平台及两个研究维度；第4章对测量设备进行配置；第5章与第6章分别报告两条研究路径；第7章探讨有效性验证所面临的挑战；第8章总结后记，阐述后续项目应采取的不同做法。

2 背景与相关工作

2.1 Alpha 研究——通俗易懂

本报告采用了一个在量化金融领域中为标准、而在其他领域则较为生僻的词汇集；本节将对该词汇集进行一次定义，从而使报告其余部分无需依赖金融背景即可阅读。*Alpha*因子是一种在每个

时点为每项资产分配一个评分的规则，其构建方式为：

即评分较高的资产随后应跑赢评分较低的资产。这一预测关注的是在固定资产集合内的*相对*表现，而非价格水平：研究对象是*横截面*——即在某一时间点观测到的所有资产——以及当前的资产评分排名是否能够预示随后数日内实际实现的收益排名。一个仅基于资产自身历史数据（如五日价格反转、异常交易量）对资产进行评分的因素，在本报告术语中称为*孤立因子*；而*关系因子*则可能分析资产在资产网络中的邻接关系。

在第5章与第6章的每一页中均出现两次相关测量指标。*信息系数*（IC）是指在某一时间点计算出的得分与随后实现的收益之间的相关性；本报告采用秩形式（*秩-IC*，即斯皮尔曼相关系数），对每个时间点进行横截面计算后取平均值。*秩-IC*为1表示完全排序；为0则表示完全无预测能力。实际因子的预测效力较弱——本报告中样本外表现最佳的综合因子的*秩-IC*约为0.015（第5章）——此类强度的因子仅在组合配置时才具有实用价值，因此该领域构建了大量此类因子。*夏普比率*则将因子作为一种交易策略进行评估：买入排名靠前的资产，做空排名靠后的资产（构建一个长空组合，其对整体市场方向的敞口在很大程度上相互抵消），然后将该策略的平均收益率除以其年化波动率。这两个数字可能不一致，因为排名靠前并不能反映胜场与负场数量的实际规模；第6章正是将这种不一致之处转化为本报告的主要发现之一。

“Alpha”这一术语本身——该术语源自本报告沿袭自Kakushadze的从业者常用公式集[1]——所指的内容并不深奥，仅指一种预期能承载超越被动市场敞口之预测信息的信号。下文中的任何陈述均不依赖于更严格的定义。

2.2 Cross-sectional alpha research

The equity track’s island factors follow the formulaic-alpha idiom that Kakushadze documented from industry practice: short arithmetic expressions over price and volume, evaluated per stock and ranked across the cross-section each day [1]. Individually weak and mutually correlated, such alphas are meant to be combined, and the interesting research questions live in the combination step. Machine learning has a measured track record here: Gu, Kelly and Xiu showed on six decades of US data that flexible models — trees and shallow networks in particular — roughly double the predictive R^2 of linear benchmarks for cross-sectional returns, driven largely by interaction effects that linear factor models cannot express [3]. The same literature is emphatic about how easily such gains evaporate under sloppy evaluation. López de Prado catalogues the failure modes — label overlap, missing embargoes, selection on test data, backtest overfitting — and the purged, embargoed validation splits he advocates [4] are the direct ancestors of the split protocol in Section 4.3. This report treats that methodological literature as load-bearing: two of the energy track’s three artifacts (Section 6.2) are textbook instances of exactly the pathologies it names.

2.3 Graph neural networks and attention

Graph neural networks compute node representations by aggregating information from graph neighbourhoods. In the graph convolutional network of Kipf and Welling, the aggregation weights are fixed by the graph’s degree structure [5]; graph attention networks replace them with weights computed from the node features themselves, so the model learns which neighbours matter for each node [2]. Brody, Alon and Yahav later showed that the original formulation computes a restricted, effectively static form of attention and proposed GATv2, which reorders the operations to make the attention genuinely conditional on the query node [6]. Both operators are available off the shelf in PyTorch Geometric [7]; the alpha track uses the original operator, the forecasting work the GATv2 variant (Section 4.4).

Applying relational models to stock prediction is an established idea. Feng et al. combine a temporal encoder with relational graphs over company links and learn to rank stocks by return [8], and much subsequent work follows the same pattern: propose an architecture, report that it beats non-relational baselines. This project asks a deliberately narrower question. Rather than demonstrating that some graph-augmented architecture outperforms, it holds the information set fixed — the same island alphas every baseline consumes — and measures the marginal value of each ingredient separately: the graph via an unlearned uniform-propagation anchor, and learned attention via the gap between the GAT and that anchor (Section 4.6). The price of the narrower question is weaker headline numbers; the payoff is that when the equity track reports a positive attention margin in thirty of thirty runs (Chapter 5), the claim isolates a mechanism rather than a model.

2.4 Electricity price forecasting

Day-ahead electricity prices are set in a daily auction and driven by fundamentals — load, wind and solar in-feed, fuel prices — with properties equity researchers rarely face: strong diurnal seasonality, negative prices, and fat-tailed scarcity spikes. Weron’s review maps the field’s method families and stresses the role of these fundamentals and of rigorous out-of-sample evaluation [9]; Lago et al. sharpen the evaluation point for the deep-learning era, offering an open benchmark and best practices — among them, always comparing against naive and well-tuned simple baselines — precisely because the literature is rich in models that beat only weak straw men [10]. The forecast-skill ladder of Section 4.7, with persistence pinned at zero skill and a ridge on fundamentals as the serious no-graph baseline, is this project’s implementation of that advice. Its energy-specific contribution is not a new forecasting architecture but a controlled measurement of one factor the standard benchmarks do not isolate: whether the physical interconnector topology itself

2.2 横断面 α 研究

权益赛道中的岛屿因子遵循Kakushadze从行业实践中归纳出的公式化阿尔法计算方法：基于价格与成交量构建的简短算术表达式，逐只股票进行计算，并每日对横截面数据进行排序[1]。这些阿尔法因子个案表现疲软且相互关联，因此旨在进行组合构建，而引人入胜的研究问题正存在于组合构建这一环节。机器学习在这一领域拥有经得起检验的应用记录：Gu、Kelly与Xiu利用美国六十年的市场数据表明，灵活模型——尤其是决策树与浅层神经网络——能使线性基准模型对横截面收益率的预测 R^2 值提高约一倍，这主要得益于线性因子模型无法体现的交互效应[3]。同一批文献也强调了此类收益在粗放式评估下极易消散的现象。Lopez de Prado系统梳理了各类评估失效模式——标签重叠、embargo 信息缺失、基于测试数据的筛选、回测过拟合——而他倡导的经过清洗且排除 embargo 信息的验证集划分方式[4]，正是第4.3节所述划分方案的直接前身。本报告将相关方法文献视为承重部分：在“能量轨迹”所涉及的三项研究对象中，有两项（第6.2节）正是该文献所指出的各类病理现象的典型例证。

2.3 图神经网络与注意力机制

图神经网络通过聚合来自图邻域的信息来计算节点表示。在 Kipf 和 Welling 提出的图卷积网络中，聚合权重由图的度结构决定[5]；而图注意力网络则用根据节点特征本身计算出的权重来替代这些聚合权重，从而使模型能够学习出对每个节点而言哪些邻节点具有重要意义[2]。Brody、Alon 和 Yahav 后来证明，原始公式计算出的是一种受限的、本质上静态的注意力机制，并提出了 GATv2——该模型重新排列了运算顺序，使得注意力机制真正地依赖于查询节点[6]。这两种运算符均可在 PyTorch Geometric[7]中直接使用；Alpha Track 采用原始运算符，而预测相关研究则采用 GATv2 变体（第4.4节）。

将关系型模型应用于股票预测是一个成熟的思路。Feng 等人将时序编码器与基于公司关联关系构建的关系图相结合，通过股票收益率[8]对股票进行排序学习；随后的大量研究也遵循了相同的模式：提出一种架构，并报告该架构优于非关系型基线模型。本项目则提出了一个意图上更为狭窄的研究问题：它并非旨在证明某种基于图结构增强的架构具有更优性能，而是将信息集保持固定——即所有基线模型均使用的同一组“岛屿阿尔法”参数——并分别衡量每个组成部分的边际价值：其中，图结构通过一个未学习的均匀传播锚点进行衡量；而学习到的注意力机制则通过 GAT 与该锚点之间的差异来衡量（第4.6节）。这种更窄研究范围所带来的代价是所得出的总体数值相对较弱；而其回报在于：当“股票基准模型”在全部 30 次实验中均报告出正的注意力边际值时（第5章），这一结论所揭示的是一种机制，而非某个特定的模型。

2.4 电价预测

日前电力价格通过每日拍卖方式确定，其形成受负荷、风电与光伏并网量、燃料价格等基本面因素驱动——而房地产领域权益研究人员则较少遇到以下情况：强烈的昼夜季节性波动、负电价以及厚尾型稀缺性峰值。Weron 的综述梳理了该领域的各类定价方法体系，并强调了这些基本面因素以及严谨的样本外评估在其中所起的作用[9]；Lago 等人则针对深度学习时代优化了评估视角，提供了一个公开的基准体系及最佳实践——其中一项重要实践是始终与朴素模型及经过充分调校的简单基线模型进行对比。——正是由于现有文献中充斥着诸多仅能胜过低效“稻草人”模型的预测模型[10]。本项目对第4.7节中提出的“预测技能梯度法”进行了实践应用：该方法将预测持续性设定为零技能，并以基本面因素作为严肃的无图基准线。该项目在能源领域所作的贡献并非提出一种全新的预测架构，而是对标准基准测试所未分离的一个因素进行了一项受控的测量——即：物理互联电网拓扑结构本身是否能提升预测技

adds forecast skill, at the node level and — where the answer turns out to be strongest — for cross-border spreads (Section 6.6).

2.5 Where this report sits

Against this background, three commitments distinguish this report from the related work above, which rarely combines them. Every relational claim is measured against matched no-learning anchors on identical inputs, so a positive result isolates an ingredient rather than crediting a whole architecture. One GAT kernel and one evaluation harness are carried across two structurally different graphs — an estimated correlation backbone over equities and a physical transmission network over bidding zones — with the negative controls travelling to both. And the honesty standard of the methodological literature is applied without exception: negative results (a failed value-added gate, a money-losing energy strategy, a null congestion hypothesis) are reported as findings, because the diagnostic arc that produced them is the part of this work most likely to help a reader.

3 Platform Architecture

The research of Chapters 5 and 6 runs on a data-engineering platform that was prior work to this project; the project’s architectural contribution is the pair of seams that let relational factors join that platform without forking it, and the loop that carries research outputs back into the warehouse. This chapter describes the platform in the depth the later chapters rely on — enough to make every named component findable in the repository — and no further.

3.1 The layered platform

The platform follows a conventional analytics layering. Ingestion fetches daily equity bars (Yahoo Finance), deterministic synthetic prices for offline work, and the ENTSO-E power-market series used in Chapter 6: day-ahead prices, load and renewable forecasts, realised load, generation mix, and physical cross-border flows with transfer capacities [11]. Raw and processed data land as Parquet files in a local data lake. The warehouse is DuckDB [12], chosen for zero-infrastructure local reproducibility — anyone with the repository can rebuild every table. Transformations are split by nature: factor mathematics lives in Python, while warehouse marts are dbt models [13] so that derived tables are declarative, tested, and documented. A Kestra flow orchestrates the daily equity and hourly energy pipelines, and a Streamlit dashboard is the read-only application layer.

One boundary is stated in the architecture documentation and repeated here: the platform is a research and portfolio-demonstration system. It places no orders, manages no broker state, and offers no investment advice.

3.2 Two seams instead of a fork

The capstone’s design problem was to add a second *family* of factors — relational ones — to a platform whose registries, apply step, backtests, and marts all assumed island factors. The naive solution is a parallel pipeline; the project’s solution is two narrow interfaces, fixed in architecture decision records before implementation.

The first seam is the propagator: one snapshot’s node features and topology in, one factor value per node out. Both the unlearned baseline (uniform neighbour averaging) and the trained GAT are adapters on this single interface. The seam is deliberately per-snapshot and transform-only — training happens elsewhere, weights are bound at construction — which keeps it a pure, testable function and keeps attention behind the interface rather than in it. The design consequence

能（无论是在节点层面，还是——在答案最为显著的情况下——对于跨境价差而言，详见第6.6节）。

2.5 本报告位置

在此背景下，本报告通过三项核心承诺与上述相关研究区分开来——后者鲜少将这些承诺有机结合。本报告中的每项关系性命题均针对在相同输入条件下匹配的“无学习基准”进行检验；因此，阳性结果旨在揭示某个特定要素，而非为整个架构正名。本研究采用一个GAT核函数和一个评估框架，将其应用于两个结构迥异的图模型：一个是基于股票市场的估计相关性骨干网络，另一个是基于投标区的物理传输网络；而阴性对照则同时应用于这两个图模型。此外，本研究无一例外地采用了方法论文献中所倡导的“诚实性”准则：阴性结果（如增值门控机制失效、亏损型能源策略、无效拥塞假设等）均被作为独立研究发现予以报告，因为产生这些结果的诊断性分析路径正是本研究中最有助于读者理解的部分。

3 平台架构

第5章与第6章中的研究基于一个数据工程平台——该平台是本项目前期开展的工作；本项目在架构上的创新在于引入了一对“接缝”机制，使关系型因素能够无缝连接该平台而无需对其进行拆分，以及一条将研究结果回传至数据仓库的循环路径。本章对该平台进行详细描述（其详尽程度足以确保所有被命名的组件在代码库中均可被定位），但不再作进一步阐述。

3.1 分层平台

该平台采用传统的分析层架构。数据采集涵盖每日股票数据条（来自Yahoo Finance）、用于离线计算的确定性合成电价，以及第6章中使用的ENTSO-E电力市场序列——包括日前电价、负荷与可再生能源预测、实际负荷、发电结构及具有输电容量的实物跨境流[11]。原始数据与处理后数据均以Parquet文件形式存储于本地数据湖中。数据仓库采用DuckDB[12]，此举旨在实现零基础设施的本地可复现性——任何拥有该存储库的用户均可重建所有数据表。数据转换工作按性质划分：因子数学运算在Python中实现，而数据仓库集市则采用dbt模型[13]，从而使派生表具备声明式、可测试及可文档化的特性。一个Kestra流程负责协调每日股票数据与每小时能源数据的处理管线，而Streamlit仪表板则作为只读的应用层。

架构文档中已明确说明的一项边界在此重申：该平台是一个用于科研及投资组合演示的系统；它不接受订单、不管理经纪商状态，也不提供任何投资建议。

3.2 采用两条缝线而非叉形缝合线

本次顶点项目的设计任务是在一个平台基础上新增第二个家族的因子（即关系型因子）；该平台原有的注册表、应用步骤、回测模块及投资组合均基于“孤岛式”因子设计。一种简单的解决方案是采用并行处理流水线；而本项目采用的解决方案则是定义两个窄接口，并在实施前将其固化于架构决策记录中。

首条缝线即为传播器：输入为单次快照的节点特征与拓扑结构，输出为每个节点对应的一个因子值。无论是未训练的基线模型（均匀邻域平均法）还是已训练的GAT，均作为该单一接口上的适配器。该缝线被刻意设计为针对单次快照且仅涉及变换——训练过程在其他环节完成，权重在构建时即已绑定——这使得它成为一项纯粹且可验证的函数，并将注意力机制置于接口之外而非接口内部。这一设

carried through both result chapters: because baseline and GAT share one seam, the attention A/B of Section 4.6 is a one-line adapter swap, not a comparison across two code paths that might differ in untracked ways. A characterisation test pins the equivalence: the uniform propagator on a fully-connected topology must reproduce the platform’s pre-existing cross-market mean exactly.

The second seam unifies factor declaration. The two tracks had grown two factor idioms — equity as metadata plus a compute callable, energy as metadata interpreted by a monolithic imperative function. One `Factor` contract over the canonical *(time, entity)* panel, plus a `FactorProvider` interface for anything that yields factors, absorbed both: the energy legacy path is wrapped as one provider, and the graph factor is simply another, whose compute closes over a propagator and a topology source and runs the per-snapshot loop internally. At the registry and apply layer, island and relational factors are indistinguishable — which is precisely what makes the four-gate comparison of Section 4.5 a fair one, since every factor reaches the evaluation through the same code.

3.3 From research run to warehouse mart

Research results do not live in notebooks. A persistence step flattens a GAT run into four warehouse tables — the factor panel (composite, anchors, and island alphas per date and symbol), per-alpha diagnostics, the four gates as a one-row report, and the attention A/B as another — and dbt models build the presentation layer on top: a staging view over the panel, a tiered factor-versus-baseline mart, and a one-row scorecard joining gates to the A/B. The dashboard reads the marts, not the research code. The practical effect showed up during this report’s own preparation: rebuilding the marts end-to-end (21 dbt models and tests on the equity warehouse, 12 on the energy warehouse) is a single command per project, and the full test suite — 115 tests, of which 55 belong to the capstone modules — is the reproducibility contract for everything the previous chapters claimed. The leakage-critical pieces (labels, splits, rank-IC) are deliberately pure pandas so they are testable without the GNN dependency stack.

4 Methodology

Both research tracks run through one fixed measurement apparatus. This chapter defines it once: the data representation (Section 4.1), the supervision label (Section 4.2), the split protocol and its leakage controls (Section 4.3), the model and training objective (Section 4.4), the four research gates (Section 4.5), and the attention A/B that carries the core thesis test (Section 4.6). Section 4.7 adds the forecast-skill apparatus used after the energy track was reframed from alpha research to forecasting. Every experiment in Chapters 5 and 6 reports against this apparatus without modification; where a design choice was made in response to an observed failure, the chapter that reports the failure also documents the change.

4.1 The canonical panel

All data enters the pipeline as a panel indexed by *(time, entity)*: *(date, symbol)* on the equity track, *(timestamp, bidding zone)* on the energy track. The entities are the nodes a graph later connects. A *snapshot* is the cross-section of one timestamp — the unit that graph propagation operates on; no operation ever sees the panel as a whole time series.

Node features are deliberately not raw market data. Each track already maintains a registry of island factors — alphas that score a node from its own history alone, ten on the equity track and eight on the energy track. Their cross-sectionally ranked values form the feature matrix. This choice pins the research question: any value the graph model adds must come from relating nodes, because the per-node information is the same information the island baselines consume.

计思路贯穿于两章结果部分：由于基线模型与GAT共享同一条缝线，因此第4.6节中的注意力机制A/B仅需要通过一行适配器代码进行替换，而非对两条可能以非追踪方式存在差异的代码路径进行比较。一项特性验证可确认其等价性：在全连接拓扑结构上应用的均匀传播器必须精确复现平台原有的跨市场均值。

第二条连接线实现了因子声明的统一。这两条数据流此前各自形成了两种因子用法：一种是将“股权”作为元数据结合一个可调用的计算函数；另一种是将“能源”作为由单体命令式函数解释的元数据。基于标准（*时间*、*实体*）面板构建的 Factor 合约，以及为所有可生成因子的组件提供的 FactorProvider 接口，共同整合了上述两种用法：遗留的“能源”处理路径被封装为一个 Provider，而图结构因子则作为另一个独立的 Provider——其计算过程围绕一个传播器和一个拓扑源展开，并在内部运行针对每个快照的循环。在注册表与应用层，岛屿因子与关系型因子无法区分——这正是第4.5节中对“四门控”模型进行公平比较的原因，因为所有因子均通过相同的代码路径进入评估阶段。

3.3 从研发阶段到仓库管理系统

研究结果并不存储于笔记本中；通过执行持久化步骤，可将一次 GAT 运行结果扁平化为四个仓库表——包括因子面板（按日期和符号分别包含复合因子、锚因子及岛屿因子）、各因子诊断结果、将四个门控逻辑整合为单行报告，以及注意力 A/B 表。

— DBT 模型则构建了上层的呈现层：包括对面板的预览视图、分层的因子与基线数据集市，以及将各类门控逻辑与 A/B 测试连接起来的单行记分卡。该仪表板直接读取数据集市，而非直接调用研究代码。这一做法的实际效果在本报告的编写过程中得到了验证：对数据集市进行端到端的重建（在权益资产仓库中运行 21 个 DBT 模型及测试，在能源资产仓库中运行 12 个）只需为每个项目执行一次单条命令；而完整的测试套件（共 115 个测试，其中 55 个属于 Capstone 模块）则构成了对前文各章节所述所有内容可复现性的保障。对于对数据泄露至关重要的组件（如标签、数据集划分、Rank-IC），我们特意采用纯 Pandas 代码实现，以便在无需依赖 GNN 相关技术栈的情况下对其进行测试。

4 研究方法

两条研究路径均贯穿于同一套固定测量装置。本章对此进行统一定义：数据表示（第4.1节）、监督标签（第4.2节）、分割协议及其泄漏控制（第4.3节）、模型与训练目标（第4.4节）、四个研究门（第4.5节）以及承载核心论点检验的注意力机制A/B（第4.6节）。第4.7节补充了在将能源研究路径从阿尔法研究重新定义为预测研究后所采用的预测技能装置。第5章与第6章中的所有实验均直接参照该装置进行报告；若因观察到的失败而做出设计调整，则报告该失败的章节亦会记录此项调整。

4.1 标准检测组

所有数据均以由（*时间*、*实体*）：（*日期*、*代码*）（对应股票交易通道）或（*时间戳*、*投标区*）（对应能源交易通道）索引的面板形式输入到数据流中。这些实体即为后续图结构中所连接的节点。*快照*是指针对单个时间戳所取的截面——这是图结构传播所作用的单元；任何操作均不会将该面板视为一个完整的时序序列。

节点特征并非直接取自原始市场数据。每条交易轨道均已建立一套岛屿因子登记册——这些 α 系数仅基于节点自身的历史数据对节点进行评分：股票交易轨道的评分上限为10分，能源交易轨道的评分上限为8分。这些跨期排序后的数值共同构成特征矩阵。这一设计明确了研究问题：图模型所添加的任何数值都必须源于节点间的关联关系，因为每个节点所包含的信息与岛屿基准模型所使用的信息完

Concretely, the equity ten are functions of each stock’s own prices and volumes alone (reversal, medium-horizon momentum, intraday range position, volume shocks, overnight-gap behaviour); nothing exogenous enters that track. The energy eight mix each zone’s price dynamics with its published fundamentals, such as residual load, wind and solar forecast terms, demand surprise, and the gas spark spread. Which signals enter matters less than the rule that they enter everywhere: island baselines, anchors, and GAT consume this identical matrix, so a relational model can hold no information advantage over the baselines it is measured against. Missing values are filled with the per-snapshot median, which in rank space is the no-information position; a node with no data is indistinguishable from a node ranked exactly in the middle.

4.2 Supervision label

The label is the forward return over a fixed horizon k , standardised per snapshot so that the model learns to rank the cross-section rather than to predict absolute magnitudes. On the equity track it is $r_{t,k} = p_{t+k}/p_t - 1$ with $k = 5$ trading days, z-scored within each snapshot. On the energy track it is $r_{t,k} = (p_{t+k} - p_t) / \max(|p_t|, 20)$ with $k = 24$ hours, clipped to ± 0.8 and z-scored; the denominator floor is unavoidable because European power prices cross zero and turn negative, so a plain relative return divides by numbers arbitrarily close to zero.

The label is supervision only. It is computed from prices at $t + k$ and never appears as a feature; the last k snapshots of each entity have no label and are excluded from training. The energy clip, introduced for label stability, later turned out to have a second life as an *evaluation* artifact — Chapter 6 reports how clipping the evaluation return concealed tail losses, and why the shipped evaluation is unclipped.

4.3 Split protocol and leakage controls

Leakage was treated as the primary methodological risk from the start, and the controls are structural rather than procedural: they are asserted in code, not remembered by the experimenter.

A single in-sample/out-of-sample boundary is placed at 70% of the timestamps. That one date is computed once and passed explicitly to every consumer: graph construction builds edges only from data strictly before it, model selection sees only data inside it, and the gate evaluation scores only data after it. The alternative — letting each component derive its own boundary — failed quietly during development: three components each assumed 70% and happened to agree, so changing any one ratio would have leaked without any test failing. Centralising the date turned that silent coupling into an explicit contract.

Within the in-sample region the layout is

$$\text{train} \mid \text{embargo}(k) \mid \text{valid} \mid \text{embargo}(k) \mid \text{OOS},$$

with the validation block embargoed on both sides. The leading embargo is standard practice: the last training labels reach k steps forward and must not overlap the first validation features [4]. The trailing embargo is easy to omit and was nearly omitted here: validation labels also reach k steps forward, so without it, best-epoch selection would be scored partly on returns that belong to the out-of-sample window. A dedicated assertion (`is_constrained_split`) rejects any split violating this layout.

Two automated controls argue that the pipeline as a whole produces no false positives. A negative control shuffles the labels and asserts that validation IC collapses to zero; a positive control plants a synthetic signal and asserts that training recovers it. Both run in the test suite on every change to the leakage-critical code, which is deliberately pure pandas — testable without a GPU or the torch dependency.

全相同。具体而言，股票交易轨道的10项系数仅是各股票自身价格与成交量的函数（包括反转信号、中短期动量信号、日内波动幅度定位、成交量冲击、隔夜缺口行为等）；该轨道不引入任何外生变量。而能源交易轨道的8项系数则将各区域的价格动态特征与其公布的基本面数据相结合，例如残余负荷、风能与太阳能预测项、需求意外值以及天然气点火价差。这些信号本身所具有的重要性，远不及“所有模型均需使用该特征矩阵”这一规则：岛屿基准模型、锚定模型及GAT均采用相同的特征矩阵，因此，对于被测模型而言，关系型模型并不具备相对于基准模型的信息优势。缺失值采用每个快照对应的中位数进行填充；在排序空间中，该中位数对应于“无信息”状态；一个无数据的节点与恰好排在中间位置的节点无法区分。

4.2 监督标签

该标签表示在固定时间窗 k 内的正向回报率，经每期快照标准化处理，使模型学习对截面进行排序而非预测绝对值。在股票交易序列中，其计算公式为 $rt, k=pt+k/pt_1$ ，其中 $k=5$ 个交易日，并在每个快照内进行 z 分数标准化。在能源交易序列中，其计算公式为 $rt, k=(pt+k_pt)/\max(|pt|, 20)$ ，其中 $k=24$ 小时，截断值为 ± 0.8 并进行 z 分数标准化；分母下限是不可避免的，因为欧洲电力价格会穿越零点并转为负值，若采用简单的相对回报率计算，将除以任意接近零的数值。

该标签仅用于监督学习。它根据 $t+k$ 时刻的价格计算得出，但从未作为特征出现；每个实体的最后 k 个快照均无标签，因此不参与训练。为确保标签稳定性而引入的“能量截断”机制，后来被证明还可作为一项评估人工指标——第6章将阐述对评估回报值进行截断如何掩盖尾部损失，以及为何最终交付的评估结果未进行截断。

4.3 分段协议与泄漏控制

从一开始，泄漏问题就被视为主要的方法学风险；而对照组属于结构层面而非程序层面——这些对照组是通过代码实现的，而非由实验者手动记忆。

在样本内/样本外边界处设定一个单一的分界点，该点位于时间戳序列的70%位置。该分界点仅计算一次，并显式地传递给所有组件：图构建阶段仅基于其之前的所有数据构建边；模型选择阶段仅查看其内部的数据；而门控评估阶段则仅对其之后的数据进行评分。另一种方案——允许每个组件自行推导其分界点——在开发阶段并未引发问题：三个组件均默认采用70%的分界点，且结果恰好一致；因此，即使更改其中任何一个分界点的比例，也不会导致任何测试失败。通过将分界点集中化处理，这一原本隐性的耦合关系被转化为一项明确定义的契约。

在样本区域内，布局为：

训练 | 禁运 (k) | 有效 | 禁运 (k) | OOS

其中验证集与训练集均设有分隔区。前置分隔区是标准做法：最后的训练标签需覆盖 k 个时间步之后，且不得与首个验证集特征重叠[4]；而后置分隔区则易于省略，在本研究中亦几乎未采用——因为验证集标签同样覆盖 k 个时间步之后，若省略该分隔区，则最佳训练轮次的筛选结果将部分基于属于样本外窗口的回报值进行评估。一个专门的断言函数（`is_constrained_split`）可拒绝任何违反此布局规范的样本集划分方案。

两项自动化对照实验均表明，整个流水线不会产生假阳性结果。阴性对照实验通过对标签进行随机打乱，验证验证阶段的干扰项（IC）会降为零；阳性对照实验则注入一个合成信号，并验证训练阶段能够正确恢复该信号。这两项实验均在测试套件中针对所有对“泄漏敏感代码”（该代码采用纯Pandas实现——可在无需GPU或Torch依赖的情况下进行测试）的代码变更时执行。

4.4 Model and training objective

The relational model is a stacked graph attention network [2] with two layers, implemented on PyTorch Geometric [7]. (The later forecasting work of Chapter 6 uses the GATv2 variant [6]; the alpha track predates it in the codebase and keeps the original operator.) The model maps one snapshot’s node features and topology to one score per node; scores across snapshots form the relational composite factor.

Training minimises an IC loss — the negative Pearson correlation between predictions and labels within each snapshot — a differentiable stand-in for the evaluation metric, the per-snapshot rank-IC (the Spearman correlation of predictions and labels, reported throughout as the OOS IC). An MSE variant is retained for comparison; Chapter 5 reports the probe that motivated the switch, in which the IC loss recovered a planted signal far better than MSE. Model selection is by best validation IC with epoch checkpointing: the weights deployed are those of the best validation epoch, not the last one. This detail matters in practice — on the first real-data run the validation IC peaked mid-training and then decayed to roughly half its peak, so returning the final epoch would have discarded most of the signal.

Two deployment modes are evaluated throughout. *Single* fits once on the in-sample region and scores the entire out-of-sample window with frozen weights. *Walk-forward* refits at every 63-snapshot boundary through the out-of-sample window, each refit using only data whose labels predate its boundary, with the same embargoed layout per fold. Every walk-forward score therefore comes from a model that would have been trainable at that date in deployment. All headline runs are CPU; a GPU benchmark found no speedup at this graph size (about 50 nodes).

4.5 The four research gates

A relational composite is only interesting if it survives comparison against the island factors it was built from. Four gates, all scored on the out-of-sample slice, formalise that comparison (Table 4.1).

Table 4.1: The four research gates. The composite is compared against the single island alphas (ten equity, eight energy), out-of-sample.

Gate	Definition	Pass threshold
Value-added	composite OOS Sharpe vs. best single	composite > max
Consistency	IS/OOS IC same sign, magnitude retention	score ≥ 0.5
Uniqueness	max Spearman vs. each single	< 0.7
Robustness	blend: OOS coverage, IC IR, drawdown	score ≥ 0.5

The value-added gate is deliberately the strictest formulation — beating the maximum of N competitors is a higher bar than beating their mean, and a composite can be genuinely useful while failing it. Results therefore also report the beat-rate against the full set of singles, and Chapter 5 discusses the one gate this project never closed.

4.6 The attention A/B: two no-learning anchors

A GAT that beats the island factors has not yet demonstrated that *learned attention* was the ingredient that mattered; the improvement could come from neighbour averaging alone. Every run therefore carries two anchors, computed from the same inputs and, where applicable, the same topology: the *island anchor*, an equal-weight composite of the input alphas with no propagation at all; and the *uniform anchor*, the same composite averaged uniformly over the GAT’s own topology — relational, but with nothing learned. Figure 4.1 draws the arrangement: three factor constructions, one input matrix, one harness.

4.4 模型与训练目标

关系模型是一个由两层组成的堆叠图注意力网络[2]，基于 PyTorch Geometric[7]实现。（第6章中的后续预测研究采用了 GATv2 变体[6]；而 alpha track 在代码库中早于该变体出现，并保留了原始运算符。）该模型将单个快照中的节点特征与拓扑结构映射为每个节点对应的一个得分；不同快照间的这些得分共同构成关系合成因子。

训练旨在最小化 IC 损失——即每个快照中预测值与标签之间的负皮尔逊相关系数——该损失函数是用于替代评估指标“每快照秩-IC”（即预测值与标签之间的斯皮尔曼相关系数，下文统称为 OOS IC）的可微分替代指标。为便于对比，本文保留了 MSE 的一种变体；第5章报告了促使我们切换到该方法的探针实验：在该实验中，IC 损失对植入信号的恢复效果远优于 MSE。模型选择基于最佳验证集 IC 值，并采用 epoch 检查点机制：所部署的模型权重对应于验证集上表现最佳的训练轮次，而非最后一个训练轮次。这一细节在实际应用中至关重要——在首次使用真实数据进行训练时，验证集 IC 值在训练中期达到峰值，随后衰减至峰值的约一半；因此，若采用最后一个训练轮次的权重，将丢失大部分训练信号。

本文全程评估了两种部署模式：*单次拟合*模式在样本区域内进行一次拟合，随后使用冻结后的权重对整个样本外窗口进行评分；*逐步前移*模式则在样本外窗口中的每个 63 个快照边界处进行一次重新拟合，每次重新拟合仅使用标签日期早于该边界的数据，且每个折叠均采用相同的禁用数据布局。因此，每个逐步前移评分均来自一个在该部署时间点本可进行训练的模型。所有主运行均采用 CPU 执行；针对该图规模（约 50 个节点）进行的 GPU 基准测试未发现加速效果。

4.5 四个研究门控单元

关系合成仅在它经与构成它的岛屿因子进行比较后仍保持有效时才具有意义。四个在样本外切片上均获得评分的门控节点，正式化了这一比较过程（表4.1）。

表 4.1：四个研究门限。将该复合指标与单岛阿尔法值（10 个股票因子、8 个能源因子）进行样本外比较。

门	定义	通过阈值
增值	复合 OOS Sharpe 指数对比最佳单项指标	复合值 > 最大值
一致性	IS/OOS IC：符号与幅值均保持不变	分数 ≥ 0.5
唯一性	最大值 Spearman 对比每个单独值	< 0.7
稳健性	混合参数：OOS覆盖范围、IC IR、抽水	分数 ≥ 0.5

“增值门限”这一指标特意采用了最严格的设定——击败 N 位竞争对手中的任意一位，比击败他们平均表现的门槛更高；而一个组合在未能达到该门槛时，仍可能具有实际应用价值。因此，结果还列出了相对于全部单打选手的胜率；第5章则探讨了该项目始终未能突破的那道“门限”。

4.6 注意力机制 A/B：两个无学习锚点

一个优于岛屿因子的 GAT 尚未证明学习到的注意力机制是关键因素；这种性能提升也可能仅源于邻域平均法。因此，每次运行均包含两个锚点——这两个锚点由相同的输入以及（在适用情况下）相同的拓扑结构计算得出：即*岛屿锚点*（由输入 α 值按等权重组合而成，且不进行任何传播）；以及*均匀锚点*（同为上述组合，但按 GAT 自身的拓扑结构进行均匀平均——该拓扑结构具有关系性，但其中没有任何学习内容）。图4.1展示了这一结构：三种因子构建方式、一个输入矩阵和一个连接架构。

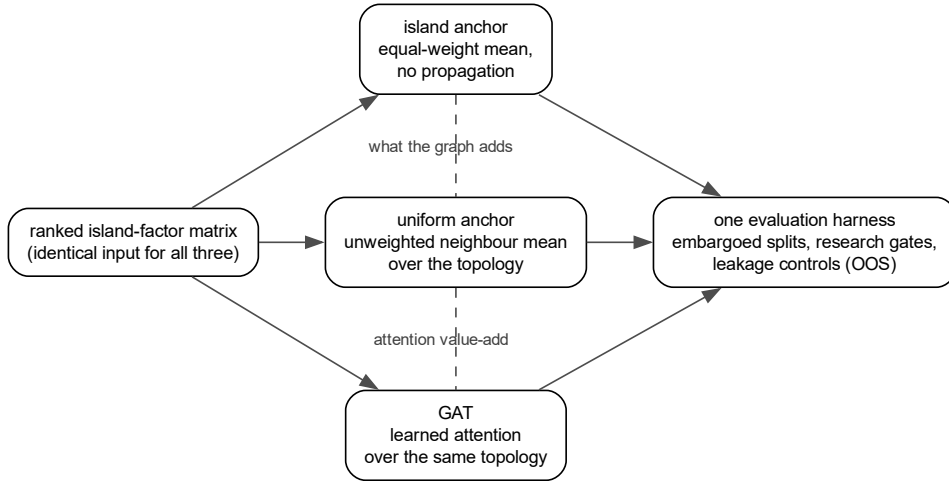


Figure 4.1: The attention A/B. All three factor constructions read the same ranked island-factor matrix and are scored by the same evaluation harness; the two dashed comparisons are the quantities this report is about — what the graph adds (island anchor versus uniform anchor) and what learned attention adds (uniform anchor versus GAT, the attention value-add).

The difference between the GAT’s out-of-sample Sharpe and the uniform anchor’s is the *attention value-add*, the quantity the core thesis stands on. It isolates the contribution of learned attention while holding the information set and the graph fixed. Where the four gates ask “is the composite a good factor?”, the A/B asks the sharper question “did learning the edge weights matter?” — and the two questions turn out to have different answers in Chapter 5.

4.7 Forecast-skill apparatus (energy reframe)

After the energy alpha attempt concluded negatively (Chapter 6), the energy question was reframed from “is there a tradeable cross-sectional signal?” to “does the physical interconnector graph improve forecast accuracy?”. The measurement apparatus changes accordingly. The metric is a skill score against persistence,

$$\text{skill} = 1 - \text{MSE} / \text{MSE}_{\text{persistence}},$$

scored out-of-sample at $k = 24$ hours, alongside MAE and the cross-sectional rank-IC. Persistence (tomorrow equals today) scores zero by construction; a model must beat carrying yesterday forward to show any skill at all.

Candidates form a ladder in which each rung isolates one ingredient: a seasonal-naive rung isolates the diurnal cycle; a per-node ridge regression on physical drivers (load, wind and solar forecasts, fuel price) isolates what a zone’s own fundamentals add; a ridge with uniform neighbour averaging isolates what the interconnector graph adds without learning; and the GAT rungs isolate what learned attention adds on top. The anchors of Section 4.6 reappear here in forecasting form — the value claim is the margin over the no-graph rung *and* over the uniform-graph rung, not the absolute skill.

Feature timing follows the day-ahead market’s information structure: load, wind, and solar forecasts for hour $t + k$ are published before t and may legitimately be used as features for $t + k$; realised quantities (actual load, generation mix) are known only up to t and enter as point-in-time anchors. The synthetic negative control carries over: synthetic zones are generated independently,

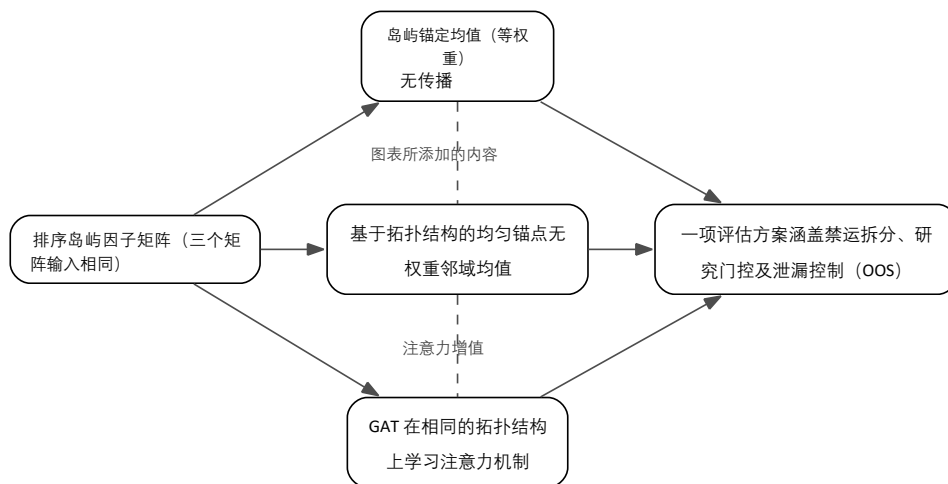


图 4.1：注意力机制 A/B。这三种因子构建方法均读取相同的排序岛屿因子矩阵，并由相同的评估工具进行评分；其中两条虚线对比线分别代表本报告所探讨的内容——即图表所体现的差异（岛屿锚点 vs 统一锚点）以及所学习到的注意力机制所带来的增量（统一锚点 vs GAT，即注意力机制带来的价值增量）。

GAT 的样本外 Sharpe 指数与均匀锚定方法所得结果之间的差异即为“注意力价值增值”——这一量值正是本文核心论点所依托的基础。该指标在保持信息集与图结构不变的情况下，独立揭示了通过学习获得的注意力机制所做出的贡献。当其他四个门控机制所探讨的是“该组合是否为一个优质因子？”时，A/B 机制则提出了一个更为精准的问题：“学习边权重是否具有重要意义？”——而在第5章中，这两个问题的答案截然不同。

4.7 预测技能装置（能量重构模块）

在“能源Alpha”研究尝试以负面结果告终后（第6章），能源问题的表述方式从“是否存在可交易的横截面信号？”转变为“物理互联电网图能否提高预测精度？”。相应的测量工具也进行了调整。该指标为一项衡量技能与持续性差异的评分。

$$\text{技能} = 1 - \text{MSE} / \text{MSE}_{\text{persistence}},$$

在 $k=24$ 小时的样本外评估中，该指标与 MAE 及横截面秩-IC 指标一同计算；由于其构建方式，持续性（即“明日即今日”）得分为零；模型必须优于“将昨日数据直接延续”这一基准，方能证明其具备任何预测能力。

候选模型构成一个层级结构：每个层级分别分离出一种要素——例如，针对季节性因素未考虑的层级可分离出日周期；基于物理驱动因子（负荷、风能与太阳能预报、燃料价格）进行节点岭回归的层级可分离出各区域自身基本面因素所贡献的部分；采用均匀邻域平均法的岭回归层级可分离出互联电网图结构所贡献的部分（无需额外学习）；而 GAT 层级则分离出通过学习获得的注意力机制所贡献的部分。第4.6节中的锚点在此以预测形式再次出现。

一价值主张是指相对于无图阶以及相对于均匀图阶所获得的边际优势，而非绝对技能水平。

特征时序性遵循日前市场信息结构：针对小时 $t+k$ 的负荷、风电及光伏发电预测数据在 t 之前发布，可合法地用作 $t+k$ 的特征；而实际观测值（实际负荷、发电组合）仅在 t 之前已知，作为时点锚点纳入分析。合成负向对照延续应用：合成区域独立生成，因此测试套件可验证在合成数据上，图

so the test suite asserts that the graph rung does not beat the no-graph rung on synthetic data. Any measured graph lift can then only come from real cross-zonal coupling.

5 Equity Track: GAT Relational Factors

The equity track asks whether propagating island-alpha information over a correlation graph produces a factor worth having, and — the sharper question, RQ1 of Chapter 1 — whether *learning* the propagation weights matters. This chapter reports the experiments in the order they constrained each other: the controls that license any positive claim (Section 5.1), the first real-data run (Section 5.2), the ablation that attributed its weaknesses (Section 5.3), the seed and device studies that set the reporting discipline (Section 5.4), hyperparameter selection under that discipline (Section 5.5), and finally what the trained attention actually does (Section 5.6). Table 5.1 states the headline up front.

Table 5.1: Equity track headline results (real yfinance data, 49 names, IC loss).

metric	value	note
Gates passed	3 / 4	Uniqueness + Consistency + Robustness pass; Value-added fails
Attention value-add	30/30 seeds positive	vs the uniform anchor (no learned attention)
Composite best-epoch validation IC	0.0756	best-by-valid-IC checkpoint selection
Composite OOS Sharpe	1.42	vs best single 2.88 (beats 9 of 10 single factors)
Graph	correlation top-k (50 names + GICS sectors)	estimated backbone (not physical)
Data	real yfinance 2021-01→2026-06	49 names; IC loss; 50 epochs

5.1 Controls before claims

No positive number in this chapter means anything unless the pipeline can be shown to find nothing where nothing exists. Three controls establish this.

First, the synthetic negative control: fifty random-walk price series run through the full pipeline — graph construction, training, gates. The gates correctly report no edge. Value-added and Consistency fail (OOS IC around -0.06 , negative Sharpe value-add), while Uniqueness and Robustness pass, as they should: they measure structural properties of a factor, not the presence of signal.

Second, the automated pair of controls that runs in the test suite on every change. Shuffling each snapshot’s labels across nodes and retraining must collapse the validation IC to zero, and it does, for both loss functions. Planting a recoverable signal — the label set equal to a standardised input feature — must let the same training loop recover it, and it does. The second half is what makes the first half meaningful: the IC collapses on shuffled labels because there is nothing to learn, not because the trainer is broken.

Third, a convergence probe for the training objective itself. On the planted-signal panel the IC loss recovers the signal essentially completely (validation IC 0.99 and above at learning rate 10^{-3}), where MSE plateaus at 0.33. The IC loss is unstable early at higher learning rates — at 5×10^{-3}

模型运行层并未优于无图模型运行层；任何测得的图模型提升效应均只能源于真实的跨区域耦合。

5 权益赛道：GAT关系因子

股权追踪部分探讨了：在相关性图上传播岛式Alpha信息是否会产生一个值得保留的因子；而——一个更为尖锐的问题，即第1章中的RQ1——学习传播权重是否至关重要。本章按实验之间相互制约的顺序报告了各项实验：为任何正向论断提供依据的对照实验（第5.1节）、首次真实数据运行实验（第5.2节）、揭示其实验局限性的消融实验（第5.3节）、确立报告规范的种子与设备研究（第5.4节）、在该规范下进行的超参数选择（第5.5节），以及最终训练后注意力机制的实际表现（第5.6节）。表5.1在开头列出了核心结论。

表 5.1：股票类投资组合总体业绩结果（基于 yfinance 实际数据，涵盖 49 只股票，计入 IC 损失）。

指标	价值	笔记
盖茨通过	3 / 4	唯一性、一致性与鲁棒性通过；增值功能未通过
关注增值	30/30种子检测结果为阳性	与均匀锚点（无学习注意力机制）对比
复合最佳训练轮次验证IC	0.0756	最佳有效IC检查点选择
复合OOS Sharpe	1.42	对比最佳单项得分 2.88（优于 10 项单项指标中的 9 项）
图	相关性 Top-K（50 个名称 + GICS 个行业）	估算主链（非物理结构）
数据	真实 YFinance：2021 年1月 → 2026年6月	49个名称；IC损失；50个训练轮次

5.1 索赔前对照组

本章中任何正数都无实际意义，除非能够证明该管道在不存在任何值的地方也未发现任何值。三项控制机制确保了这一点。

首先，合成阴性对照组：选取50条随机游走价格序列并经完整分析流程处理。——图结构构建、模型训练、门控机制。门控机制正确地报告了“无边”；“价值增值”与“一致性”两项指标未通过（OOS IC 值约为 0.06，负值的 Sharpe 价值增值），而“唯一性”与“稳健性”两项指标则通过了检验——这与预期一致：这两项指标衡量的是因子的结构性特征，而非信号的存在与否。

其次，这套自动化控制机制会在每次代码变更时在测试套件中运行。对每个快照的标签在不同节点间进行随机打乱并重新训练，必须使验证集的互信息（IC）降为零——而对于两种损失函数而言，这一目标均得以实现。引入一个可恢复的信号——即标签集等于标准化输入特征——应能使相同的训练循环将其恢复——这一目标也确实得以实现。后半部分正是赋予前半部分意义的关键：在随机打乱的标签下互信息降为零，是由于此时已无内容可学习，而非由于训练器本身存在故障。

第三，用于训练目标本身的收敛性探针。在植入信号面板上，IC损失函数基本完全恢复了信号（在学习率 10^{-3} 时，验证集IC值为 0.99 或更高），而 MSE 在 0.33 处趋于平稳。在较高学习率下（ 5×10^{-3} ）时，IC损失函数在早期阶段表现不稳定。

it bounces near zero for some forty epochs before converging — which fixed the default learning rate at 10^{-3} and the IC loss as the default objective.

5.2 First real-data run

The first run on market data used daily bars for the fifty-name universe (one name, ORCL, failed to download, leaving 49 names, 1,364 trading days and 66,836 panel rows spanning January 2021 to June 2026), the ten island alphas as node features, a static correlation graph built from data before the split date, and IC-loss training for fifty epochs.

Training behaved like the textbook picture. The training loss fell monotonically while the validation IC rose to a peak of 0.0756 at epoch 32 and then decayed to 0.0418 by the final epoch — plain overfitting. Best-epoch checkpointing therefore deployed the epoch-32 weights; had the pipeline returned the final epoch, as an earlier version of the code silently did, the deployed model would have carried barely half the validation IC. The checkpoint fix paid for itself on its first contact with real data.

The four gates split three to one. Consistency (score 0.63), Uniqueness (maximum absolute correlation against any single alpha, 0.257) and Robustness (0.68) pass; Value-added fails, with a composite OOS Sharpe of 1.42 against 2.88 for the best single island alpha. The honest reading cuts both ways. The composite is a real, unique, consistent signal — the second-best of the eleven factors on the board, beating nine of the ten singles it was built from, and correlated with none of them above 0.26. It is genuinely new information rather than a repackaging of its inputs. But the strictest bar — beat the maximum of ten competitors — stayed open here and, as the rest of this chapter shows, never closed.

The run also posed the question that drove the next experiment: the validation IC of 0.076 decayed to an OOS IC of 0.007 over the roughly 1.6-year OOS window. Was the frozen correlation graph going stale, or the frozen model weights?

5.3 Attributing the decay: graph versus model

A 2×2 ablation crossed the graph axis (static, frozen at the split date, versus dynamic, rebuilt per snapshot from a trailing window) with the retraining axis (single fit versus walk-forward refits every 63 snapshots). All four cells used one shared data fetch (all fifty names this time, 68,200 rows), the same split, the same seed, and carried the two no-learning anchors of Section 4.6. Table 5.2 shows the result.

Table 5.2: The graph $\times$ retraining ablation, one seed, IC loss. “Att. VA” is the attention value-add: GAT OOS Sharpe minus the uniform anchor’s. Anchors, for reference: island mean Sharpe -2.13 ; uniform anchor -1.05 (static graph) and -1.19 (dynamic).

Configuration	OOS IC	OOS Sharpe	Gates	Att. VA
static + single	-0.0051	1.04	F/F/T/T	2.09
dynamic + single	-0.0348	-1.01	F/F/T/T	0.18
static + walk-forward	0.0200	1.95	F/T/T/T	3.00
dynamic + walk-forward	0.0007	0.05	F/T/T/T	1.24

Two findings, one of them negative. Walk-forward retraining is the lever: it lifts the static-graph cell from an OOS IC of -0.005 to $+0.020$ and flips the Consistency gate to pass. The decay diagnosed in Section 5.2 was model staleness. The dynamic graph, by contrast, hurts in both arms — per-snapshot correlation estimates from a trailing window are noisy, and the frozen train-period graph evidently acts as a regulariser. This negative result is worth having: the intuitively appealing refinement (“keep the graph fresh”) makes things worse as implemented, and

该参数在约 40 个训练周期内保持在接近零的水平，随后收敛——此设置将默认学习率设为 10^{-3} ，并将 IC 损失函数设为默认优化目标。

5.2 首次真实数据运行

首次基于市场数据的运行采用了以下参数：针对包含50只股票的样本集（其中一只股票 ORCL 未能成功下载，因此最终包含49只股票、1,364个交易日及66,836行面板数据，时间跨度为2021年1月至2026年6月）；将十项“岛屿阿尔法”指标作为节点特征；使用基于分拆日期前数据构建的静态相关性图；并采用IC-loss训练方法进行50个训练周期的训练。

训练过程与教科书中的示意图一致：训练损失值呈单调下降趋势，而验证集的 IC 值在第 32 个训练周期时升至峰值 0.0756，随后在最终训练周期时降至 0.0418。

—简单的过拟合问题。因此，采用“最佳训练轮次检查点”策略时，部署的模型采用了第 32 训练轮次的权重；若训练流程返回的是最终训练轮次的权重——正如早期版本的代码所默认的做法——那么所部署的模型所携带的验证集 IC 值将仅约为原值的一半。这一检查点修复方案在首次接触真实数据时便取得了显著成效。

四个指标的权重比为3:1。一致性（得分0.63）、独特性（与任意单一阿尔法因子的最大绝对相关系数为0.257）及稳健性（0.68）均达标；而“增值”指标则未达标——其综合OOS夏普比率仅为1.42，而最佳单一岛屿阿尔法因子的该比率高达2.88。这一结论具有双向适用性：该综合指标实为一个真实、独特且具一致性的信号——在全部11个因子中位列第二，击败了其所构建的10个单一因子中的9个，且与这些因子的相关系数均未超过0.26。这实质上是全新的信息，而非对输入数据的简单重新包装。然而，最严苛的评判标准——即超越全部10个竞争对手中的最高值——在此处仍未被突破；正如本章其余部分所展示的，这一标准从未被完全超越。

此次运行还引出了驱动后续实验的关键问题：在约 1.6 年的 OOS 期间内，验证集 IC 从 0.076 下降至 OOS IC 的 0.007。是冻结的相关性图谱变得过时了，还是冻结的模型权重本身已失效？

5.3 衰减原因分析：图示与模型对比

一个×2的消融实验将图轴（静态——在分割日期时冻结；versus 动态——根据后置窗口的快照进行重建）与重新训练轴（单次拟合 versus 每 63 个快照进行一次前向走步重新拟合）进行交叉对比。所有四个实验均采用相同的共享数据获取方式（本次涉及全部 50 个名称，共 68,200 行）、相同的数据集分割方式、相同的随机数种子，并应用了第4.6节中所述的两个无学习锚点。表5.2展示了实验结果。

表 5.2：图×重新训练消融实验（单个种子，IC 损失）。“Att. VA”表示注意力值增益：

GAT OOS Sharpe比率（扣除均匀锚定因子后的值）。锚定因子参考值：岛屿平均Sharpe比率为 2.13；均匀锚定因子为 1.05（静态图表）及 1.19（动态图表）。

配置	OOS IC	OOS Sharpe	门	Att. VA
静态 + 单个	-0.0051	1.04	F/F/T/T	2.09
动态 + 单	-0.0348	-1.01	F/F/T/T	0.18
静态 + 前向递推	0.0200	1.95	F/T/T/T	3.00
动态 + 前向递推	0.0007	0.05	F/T/T/T	1.24

两项研究结果中，有一项为负面结果。“前向走步”重训练即为触发机制：它将静态图单元从 OOS IC 值 -0.005 提升至 +0.020，并使一致性门切换为“通过”状态。在第5.2节中诊断出的衰减现象实为模型陈旧性。相比之下，动态图在两方面均带来不利影响——来自滞后窗口的每快照相关性估计值存在噪声，而冻结的训练期图显然起到了正则化作用。这一负面结果仍具有参考价值：直观上颇具吸引力的优化方案（“保持图结构新鲜”）在实际实施中反而导致情况恶化；因此，该项目

the project kept the static graph with its mild, disclosed in-sample lookahead rather than trade OOS performance for cosmetic cleanliness.

The attention A/B, meanwhile, is positive in every cell. Both no-learning anchors are clearly negative out-of-sample; the GAT beats the uniform anchor by +0.2 to +3.0 Sharpe on the same topology with the same inputs, and its scores correlate with the uniform anchor’s at Spearman -0.29 to -0.14 — it is not learning to smooth, it is learning something else. Against the strict value-added bar the best cell still falls short (best single at 3.07; the gap narrows from -2.03 to -1.12), but the core-thesis comparison holds regardless of which cell one reads.

One warning surfaced here as well: the static+single cell read $-0.0051/1.04$ where the near-identical first run had read $+0.0066/1.42$. Run-to-run variance was clearly material, which is what the next section is about.

5.4 Variance: seeds, devices, and the reporting protocol

Five seeds per arm on the same panel put numbers on the spread. The single-fit arm scored an OOS Sharpe of 0.73 ± 0.25 (per-seed range 0.36–1.04) and the walk-forward arm 1.15 ± 0.75 (range 0.13–1.95); walk-forward’s mean advantage came with three times the variance, and a Welch t of about 1.2 at $n = 5$ per arm. The single-seed reading of the ablation — “walk-forward is the lever” — had ridden the luckiest walk-forward draw. Meanwhile the attention value-add was positive in all ten runs, spanning +1.18 to +3.00 Sharpe against a seed-independent anchor: the core claim survived the variance study; the walk-forward claim was downgraded to “improves the mean, adds variance, not significant at $n = 5$ ”.

A hardware benchmark added an unplanned second lesson. Moving training to a laptop GPU was 24–32% *slower* at this graph size — fifty nodes and ten features per snapshot is latency-bound, and kernel-launch overhead dominates — so production runs stayed on CPU. More interesting was that the same seed on a different device swung the result by the full width of the seed distribution. The seed fixes the initial weights, but dropout masks come from each device’s own random stream and parallel reductions round in a different order, so training diverges from the first step; a cross-device rerun is another draw from the same distribution, not the same experiment on new hardware.

Together these fixed the reporting protocol for everything that follows: every headline number is a mean $\pm$ standard deviation over at least five seeds on a fixed device, and what the project claims to be reproducible is the distribution, not bit-exact point estimates.

5.5 Hyperparameter selection with clean hands

The hyperparameter search was designed so that the OOS window could not influence selection even in principle. Twenty-four configurations (learning rate $\{5 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$, hidden size $\{32, 64\}$, heads $\{2, 4\}$, layers $\{1, 2\}$) ran three seeds each — 72 training runs scored exclusively on the double-embargoed validation IC, with no OOS metric computed anywhere in the grid. Only the winner then touched the OOS window, once, with fresh seeds.

The grid’s surface is flat on top: six configurations sit within validation IC 0.10–0.115, and the shipped default ranked third. The one structural regularity is depth — all six top configurations have two GAT layers. Results are therefore robust to reasonable hyperparameter choices, which is simultaneously a robustness point and a caution against reading much into tuning.

The winner’s OOS validation initially produced a finding that did not survive its own scrutiny. The validation run had silently executed on the GPU — the CUDA wheel had just been installed and the fitting routine auto-selected it — while the reference numbers it was compared against were CPU runs. Given the cross-device divergence of Section 5.4, that comparison confounded device with configuration. The code was fixed the same day (CPU is now an explicit default, CUDA opt-in) and the validation rerun on CPU. The deconfounded result: in the single arm the winner is a wash against the default (0.80 vs. 0.73 Sharpe, IC near zero in both), but in the

保留了静态图——其在样本内具有适度的前瞻性，而非以牺牲 OOS 性能为代价换取表面的整洁度。

与此同时，所有单元中的注意力权重 A/B 均为正值。两个无学习锚点在样本外数据集上均显现出负值；在相同拓扑结构和相同输入条件下，GAT 比均匀锚点高出 0.2 至 3.0 个 Sharpe 指数，且其得分与均匀锚点的得分之间的 Spearman 相关系数介于 0.29 至 0.14 之间——这并非旨在实现平滑处理，而是学习到了其他某种规律。尽管在严格的增值标准下，最佳单元仍未能达标（最佳单元单元得分为 3.07；差距从 -2.03 缩小至 -1.12），但无论分析哪个单元，其核心论点均成立。

此处还出现了一项警示：静态+单细胞读数为 $-0.0051/1.04$ ，而几乎相同的首次实验读数为 $+0.0066/1.42$ 。实验间方差显然具有重要意义，这正是下一节要探讨的内容。

5.4 方差：种子、设备及报告方案

同一组实验中每组设五颗种子，以此为分布情况标注序号。单次拟合组的 OOS Sharpe 指数为 0.73 ± 0.25 （单颗种子范围：0.36–1.04），而走步法组为 1.15 ± 0.75 （范围：0.13–1.95）；走步法组的平均优势对应的方差是前者的三倍，且在每组 $n=5$ 时，Welch t 值约为 1.2。该消融实验的单种子读数——“走步法即为操作杠杆”。

—该指标对应最幸运的“走步式”抽样序列；与此同时，在全部十组样本中，其关注度附加值均为正，其 Sharpe 指数介于 +1.18 至 +3.00 之间（以与种子无关的基准值为参照）：这一核心论点在方差检验中得以保留；而“走步式”论点则被下调为“改善均值、增加方差，在 $n=5$ 时不具统计学意义”。

一次硬件基准测试意外带来了额外的插曲：当将训练任务迁移至笔记本电脑的 GPU 上时，在当前的图规模（每个快照包含 50 个节点和 10 个特征）下，训练速度慢了 24–32%——由于该场景属于延迟受限场景，且内核调用开销占主导地位——因此生产环境中的训练仍保留在 CPU 上进行。更有趣的是，同一种子值在不同设备上运行时，结果会相差整整一个种子值分布的跨度。该种子值决定了初始权重，而 dropout 掩码则源自每台设备自身的随机数列，且并行求和操作采用不同的顺序进行，因此训练过程与第一步有所不同；这种跨设备重新运行相当于从同一分布中抽取另一个样本，而非在新硬件上重复进行同一实验。

这些规定共同确立了后续所有内容的报告规范：每个标题数值均代表在固定设备上基于至少五个种子数据计算得出的均值±标准差；而该项目所宣称的可重复性是指数据分布，而非位精确的点估计值。

5.5 纯手工超参数调优

超参数搜索的设计确保了即使在原则上，OOS 窗口也不会对参数选择产生影响。共设置了 24 种配置（学习率 $\{5 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$ 、隐藏层大小 $\{32, 64\}$ 、头数 $\{2, 4\}$ 、层数 $\{1, 2\}$ ），每种配置分别运行三次随机种子实验——共进行 72 次训练运行，这些运行仅在双重禁令验证集上进行评估，且在整个参数网格中未计算任何 OOS 指标。随后，仅对获胜模型使用新随机种子进行一次 OOS 窗口验证。

该网络的表面顶部为平坦结构：共有六种配置属于验证集 IC 0.10–0.115 范围，而出厂默认配置位列第三。唯一具有结构性规律性的特征是深度——所有六种顶部配置均包含两层 GAT 层。因此，实验结果对合理的超参数选择均具有稳健性；这一特性既体现了模型的稳健性，也提醒我们不要过度解读参数调优带来的影响。

获胜模型的 OOS 验证最初得出的一项结论在经其自身严格审查后未能成立。该验证运行是在 GPU 上后台执行的——此时 CUDA 轮盘刚刚安装，且拟合程序已自动选择了该轮盘——而用于对比的参考数据则来自 CPU 上的运行结果。鉴于第 5.4 节中提到的跨设备数据差异，该对比将设备类型与配置因素混为一谈。代码于当日得到修复（现将 CPU 设为显式默认值，CUDA 为可选配置），随后在 CPU 上重新运行了验证流程。经剔除混杂因素后的结果如下：在单臂实验中，获胜模型与默认模型表现相当（Sharpe 指数分别为 0.80 对 0.73，IC 值在两者中均接近于零）；但在走步式实验中，获胜

walk-forward arm it is better on the mean and far tighter — OOS IC 0.0148 ± 0.0043 against the default’s 0.0055 ± 0.0147 , a 2.7-fold mean with a third of the variance, all five seeds positive on both IC and Sharpe, and the IC mean 3.4 standard deviations above zero. Within this configuration, walk-forward versus single reaches significance (Welch $t \approx 3.9$ on IC, ≈ 2.4 on Sharpe at $n = 5$ per arm), upgrading the walk-forward claim to “significant in the tuned configuration, directionally consistent everywhere”.

The best known setup is therefore the static graph, IC loss, walk-forward refits, and the winner hyperparameters, at an OOS IC of 0.0148 ± 0.0043 and OOS Sharpe of 1.37 ± 0.39 . Across the three run families — seed study, GPU validation, CPU rerun — the attention value-add is positive in 30 of 30 seeded runs, spanning $+0.69$ to $+3.03$ Sharpe. That, not any single point estimate, is the equity track’s central result.

5.6 What the attention learned

The A/B says learned attention beats uniform averaging; attention analysis on a representative trained model (919,336 edge-weight rows over 1,364 dates) says how.

The attention is genuinely relational. The mean self-loop weight is 0.084 — the model places 91.6% of its attention mass on neighbours — which rules out the trivial failure mode in which attention collapses onto each node’s own features and the graph is decoration. But across a node’s roughly 12.5 neighbours the distribution is nearly uniform: normalised entropy 0.956, a top-neighbour share of 0.152 where exact uniformity would give 0.08, and a sector-homophily lift of just $+0.019$ (attention-weighted same-sector share 0.330 against a structural 0.311). The trained model is a gentle reweighting of mean pooling, not a sharp selector. This is the mechanism behind the whole chapter: small, learned tilts on top of broad pooling explain why the GAT’s edge over the uniform anchor is consistent yet modest, and why its scores are nearly uncorrelated with the anchor’s.

The coarse structure is also temporally stationary: the self/neighbour split holds at 0.91–0.92 and the homophily lift stays within $+0.01$ to $+0.03$ across the full 2021–2026 span, with no visible break at the IS/OOS boundary (Figure 5.1). The project’s original design hypothesis — that attention would adapt to macro regimes — is not supported at this aggregate level, and the report states so; whatever adaptation exists lives in per-name weights inside a stable coarse structure. Finally, the names drawing the most incoming attention are mega-caps and sector bellwethers — AMZN, NFLX, AVGO, GS, CAT, NVDA, MSFT, LIN — exactly the names the rest of the universe co-moves with, a qualitative validity check the attention passes.

5.7 Summary of the equity evidence

Learned attention over the correlation graph reliably beats unlearned propagation over the same graph with the same inputs: positive attention value-add in 30 of 30 seeded runs across two hyperparameter configurations and two devices. The resulting composite is a real, unique, consistent factor that beats nine of its ten inputs but not the best of them; the value-added gate remains the one honest failure. Dynamic graphs hurt, walk-forward retraining helps (significantly so in the tuned configuration), single-seed and single-device numbers are not to be trusted, and the attention mechanism itself is interpretable — relational, nearly uniform, stationary, and centred on economically sensible hubs.

6 Energy Track: From Alpha to Forecasting

The energy track set out to repeat the equity result on a physically grounded graph: twenty European bidding zones connected by the actual cross-border transmission network, with day-

模型在均值上表现更优且离散度显著更小——OOS IC 为 0.0148 ± 0.0043 ，而默认模型为 0.0055 ± 0.0147 ；获胜模型的均值是默认模型的 2.7 倍，而其方差仅为默认模型的三分之一；所有五个种子在 IC 和 Sharpe 指数上均呈正向；且 IC 均值比零高出 3.4 个标准差。在该配置下，走步式实验与单臂实验之间存在显著性差异（Welch $t \approx 3.9$ （IC 方面）， ≈ 2 ）。在每组 $n=5$ 的情况下，Sharpe 指数为 4；并将前向预测结论升级为“在调参配置下具有显著性，且在所有场景下均保持方向一致性”。

因此，最典型的配置为：静态图结构、IC 损失函数、前向游走重拟合法以及最优超参数；其外部验证集（OOS）IC 值为 0.0148 ± 0.0043 ，OOS Sharpe 指数为 1.37 ± 0.39 。在三个运行组别（种子研究、GPU 验证、CPU 重新运行）中，注意力机制带来的价值增值在 30 次种子运行中全部为正，其 Sharpe 指数范围在 +0.69 至 +3.03 之间。这一结果——而非任何单个点估计值——才是权益投资策略研究的核心结论。

5.6 所学的注意力

A/B 实验表明，学习到的注意力机制优于均匀平均法；通过对一个代表性训练模型（涵盖 1,364 个日期、共 919,336 行边权重数据）进行注意力分析，即可揭示其具体作用机制。

这种注意力机制本质上是基于关系的。平均自环权重为 0.084——即该模型将 91.6% 的注意力权重分配给了邻近节点——这排除了那种显而易见的故障模式（即注意力完全聚焦于每个节点自身的特征，导致整个图结构仅起到装饰作用）。然而，在一个节点所连接的约 12.5 个邻近节点范围内，注意力分布近乎均匀：归一化熵值为 0.956；其中最邻近节点所占份额为 0.152（若假设完全均匀分布则该值应为 0.08）；而“扇区同质性提升”仅为 +0.019（注意力加权的同扇区占比为 0.330，而结构化计算所得值为 0.311）。该训练后的模型是对“平均池化”方法的一种温和的重新加权处理，而非一种激进的筛选机制。这一机制正是本章论述的核心所在：在广义池化基础上引入的、经学习得到的微小偏置，解释了为何 GAT 相较于均匀锚定模型所具有的优势既具有一致性又较为温和，以及为何其得分与锚定模型的得分几乎不相关。

粗粒度结构在时间上也是平稳的：自我/邻近节点分裂系数维持在 0.91–0.92 之间，而同质性提升值在整个 2021–2026 年期间始终处于 +0.01 至 +0.03 范围内，且在 IS/OOS 边界处未见明显断层（图 5.1）。该项目最初的理论假设——即注意力会适应宏观环境——在这一总体层面并未得到支持，报告对此亦有明确说明；若存在某种适应性，则这种适应性体现在每个名称对应的权重中，且这些权重处于一个稳定的粗粒度结构之内。最后，吸引最多注意力的名称为大型股和行业领头羊。

— AMZN、NFLX、AVGO、GS、CAT、NVDA、MSFT、LIN — 这些正是宇宙中其余万物所共有的名称；这是一项注意力所进行的定性有效性验证。

5.7 股权证据摘要

在相同图结构和相同输入条件下，基于相关性图的“习得式注意力机制”表现显著优于基于同一图结构的“非习得式传播机制”：在两种超参数配置及两种设备上进行的 30 次随机种子运行中，所有运行均显示出正向注意力值的增益效应。由此得到的复合因子是一个真实、独特且一致的因子——它在 10 个输入因子中胜出 9 个，但并未胜出其中表现最佳的因子；而“增值门控”机制仍是唯一出现故障的因子。动态图结构会对模型性能产生不利影响，而采用“前向遍历重新训练”方法则能有效改善这一问题（在经过调参的配置下效果尤为显著）；单次种子运行及单设备实验所得的结果不可完全信赖；此外，该注意力机制本身具有可解释性——其特性为：基于关系、近乎均匀、平稳且以经济上合理的枢纽节点为中心。

6 能源追踪：从阿尔法模型到预测分析

该能源分析路径旨在在一个基于物理现实的图谱上复现股权分析结果：由实际跨境输电网络连接

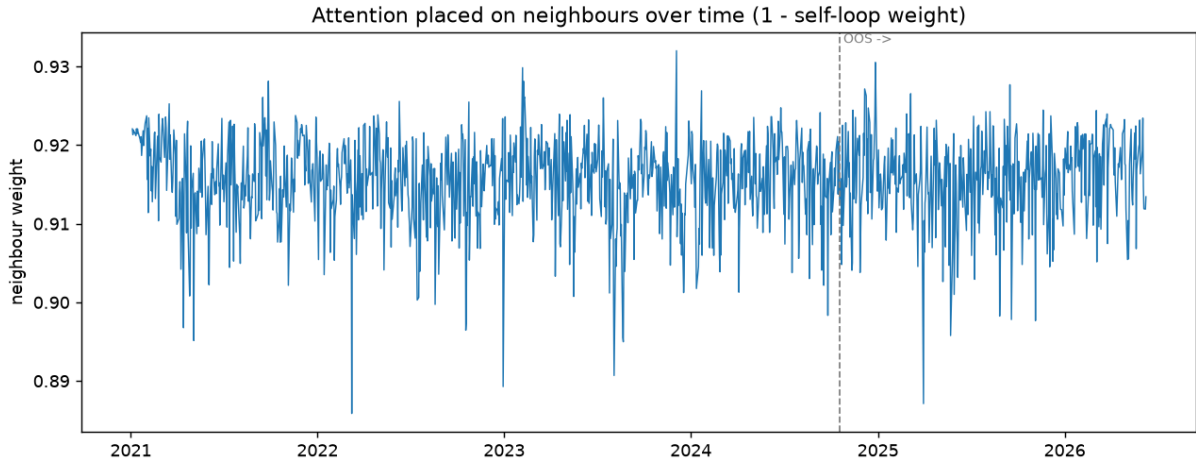


Figure 5.1: Attention mass placed on neighbours (one minus the self-loop weight) per snapshot over 2021–2026, from the archived attention analysis of the representative trained model. The share stays in a narrow band around 0.91–0.92, with no break at the dashed IS/OOS boundary — the stationarity reported in the text.

ahead spot prices from the ENTSO-E Transparency Platform [11]. Figure 6.1 shows that graph as configured in the repository — twenty zones, 38 physical borders and HVDC links, reference data rather than anything estimated from prices. The track ended somewhere more interesting than its plan. The cross-sectional alpha framing failed — not quietly, but through three nested evaluation artifacts, each of which initially produced spectacular numbers and each of which the project’s own controls caught (Sections 6.1 to 6.4). The track was then reframed as a forecasting problem, where the graph finally showed genuine, control-validated value, concentrated exactly where relational structure should matter most: on cross-border spreads (Sections 6.5 and 6.6). This chapter reports the arc in order, because the order is the contribution. Two research questions run through it: whether one kernel transfers to a physical graph without manufacturing false positives (RQ2), and whether that graph improves forecasts (RQ3).

6.1 One kernel, two graphs — and a second negative control

The dual-track design goal was engineering economy: the equity GAT, its training loop, its four gates, and its A/B anchors run unchanged on the energy panel; only the graph (physical interconnectors instead of estimated correlations), the label horizon (24 hours instead of 5 days), and the node set (bidding zones instead of symbols) differ. That goal was met literally — the same kernel functions are called by both tracks — and the first energy run doubled as a control.

On synthetic power data (2,857 hourly snapshots, 57,140 rows, generated without any cross-zonal coupling) the pipeline correctly finds nothing: OOS IC around -0.01 in both retrain arms, and, crucially, the attention value-add is *negative* (-0.74 single, -0.59 walk-forward). When there is no relational signal, the GAT is worse than uniform neighbour averaging, not better. This is the energy analogue of the equity negative control, on a structurally different graph: even a learned attention layer over a physical topology does not manufacture signal from noise.

6.2 Real data, implausible result

The live ENTSO-E run covered all twenty zones for the full year 2024 at hourly resolution — 175,200 rows, with every configured zone returning data. Real power prices immediately justified the label’s engineering: they spanned -427 to $1,896$ EUR/MWh, negative in 3.95% of hours. The pipeline ran end to end without kernel changes. The numbers it produced were an OOS IC of

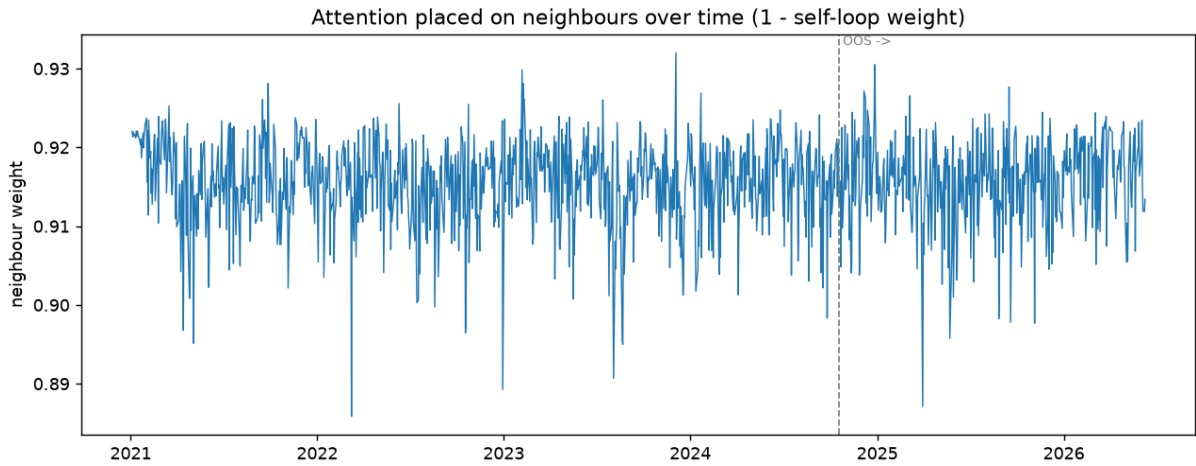


图 5.1：基于代表性训练模型的存档注意力分析结果，展示了 2021–2026 年间每个时间步长上施加于邻节点上的注意力权重（即 1 减去自环权重）；该权重值始终维持在 0.91–0.92 的狭窄区间内，且在虚线标示的 IS/OOS 边界处未出现断点——这与文中所述的平稳性特征一致。

的二十个欧洲投标区域，其日前现货价格取自 ENTSO -E透明度平台[11]。图6.1展示了该图谱在数据仓库中的配置——包含二十个区域、38条物理边界及高压直流输电链路，这些均为参考数据而非基于价格估算所得。该分析路径最终得出的结果比其规划目标更为引人注目。横截面 α 框架法未能奏效——并非悄无声息地失败，而是通过三个嵌套的评估工件实现的；每个工件最初均产生了显著的数值，而项目自身的控制机制均成功捕捉到了这些数值（参见第6.1至6.4节）。随后，该分析路径被重新界定为一个预测问题，此时该图谱最终呈现出经控制机制验证的真实价值，且该价值恰好集中于关系结构应发挥最大作用的领域：即跨境价差领域（参见第6.5和6.6节）。本章按顺序汇报这一分析路径，因为这种顺序本身即为其贡献所在。本文贯穿始终的两个研究问题为：一是某个核函数是否可无误报地迁移至物理图（RQ2）；二是该图是否能提升预测性能（RQ3）。

6.1 一个核函数、两个图——以及第二个阴性对照

双轨设计目标在于实现工程经济性：股权GAT、其训练循环、四个门控节点以及A/B锚点均在能源面板上保持不变；仅图结构（采用物理互连器而非估计相关性）、标签时间范围（24小时而非5天）以及节点集（投标区而非符号）有所不同。这一目标已得到切实实现——两条轨道调用了相同的核函数——而首次能源模拟运行同时充当了对照组。

在合成功率数据集（包含 2,857 个每小时时间序列样本，共 57,140 行数据，生成时未施加任何跨区域耦合）上，该分析流程未检测到任何异常：两个重新训练组中的 OOS IC 值均约为 $_0.01$ ；至关重要的是，注意力机制带来的附加价值为负值（单次预测： $_0.74$ ；走步式预测： $_0.59$ ）。当不存在关系性信号时，GAT 模型的表现逊于均匀邻域平均法，而非更优。这相当于在结构不同的图上应用的“权益负向对照”方法在能量层面的对应现象：即使在物理拓扑结构上部署学习型注意力层，也无法从噪声中提取出有效信号。

6.2 真实数据，却得出不合理的结论

此次实时 ENTSO -E运行覆盖了2024年全年全部20个区域，数据采样频率为每小时一次——共包含175,200行数据，每个已配置区域均返回了数据。实际电价充分验证了该标签的工程设计合理性：电价范围介于427至1,896欧元/兆瓦时之间，其中3.95%的时段为负值。该数据管道实现了端到端运行，无需对内核进行任何修改。其生成的数据构成了一套OOS集成电路。

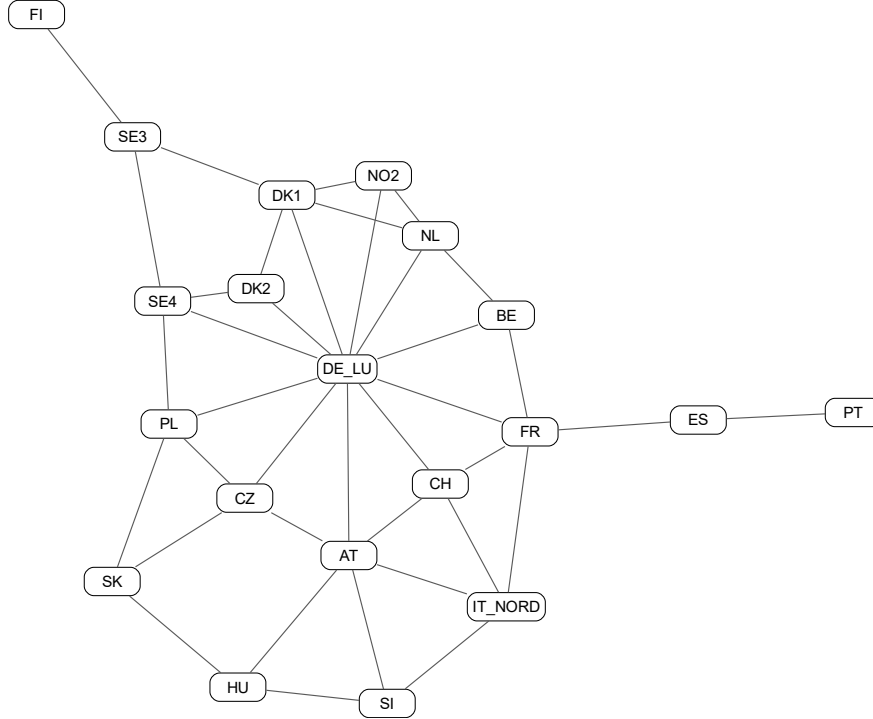


Figure 6.1: The energy graph: 20 European bidding zones and their 38 physical cross-border interconnections (land borders and HVDC cables), drawn from the repository’s reference data. Unlike the equity correlation graph, nothing here is estimated — the topology is the grid. DE_LU, with eleven interconnectors, is the network’s natural hub.

0.20, an OOS Sharpe of 8.25, a composite roughly one hundred times its best single input, and four gates out of four.

The same kernel earns a Sharpe of about 1.4 on equities. A result this much better is not a discovery; it is a symptom. Diagnosis found three compounding artifacts. First, overlapping labels: a 24-hour forward return sampled hourly gives consecutive observations that share 23 of their 24 hours — measured lag-1 autocorrelation 0.91 — while the Sharpe computation treats them as independent, understating the denominator by an order of magnitude. Second, the “signal” is trivial: a one-line predictor, the negative of the current spot price, scores a cross-sectional IC of 0.235, *higher* than the GAT’s 0.20. Whatever the model learned, it is dominated by deterministic diurnal mean-reversion. Third, the annualisation constant was the equity one (252 periods per year) applied to hourly data (8,760). On top of all three sits a market-structure lookahead: the day-ahead auction publishes all 24 hours of day D at once on $D - 1$, so an “hourly” 24-hour-ahead label is partly contemporaneous information.

The point the report wants to preserve is that the magnitude itself was the alarm. The same scepticism that made the synthetic controls a design feature flagged the real-data number as not credible, rather than publishing a Sharpe of 8.

6.3 Redesign in the day-ahead frame

The redesign moved the evaluation to the market’s actual decision frequency. The day-ahead auction clears once per day, so the snapshot became the delivery day: base-load-aggregated features from data up to day D , a label equal to the genuinely-unknown next-day base-load price change $D \rightarrow D+1$, non-overlapping by construction, annualised at 365.

The mechanical checks all came back clean. Label lag-1 autocorrelation fell from 0.91 to -0.10 ;

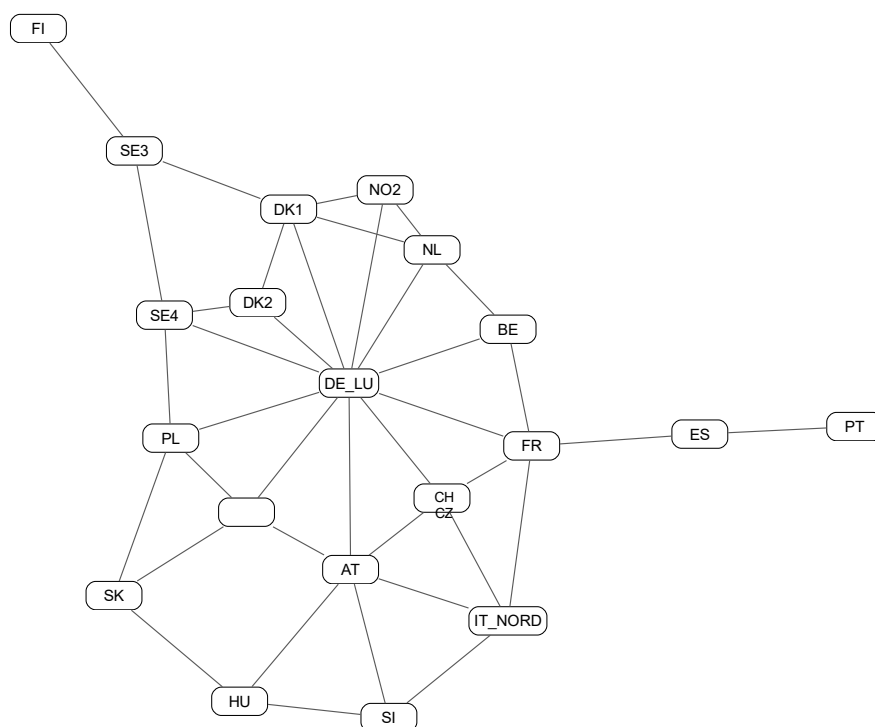


图 6.1: 能量网络图: 展示了 20 个欧洲投标区域及其 38 条实际跨境互联线路（包括陆地边界与高压直流输电电缆），数据源自该数据库的参考数据集。与股权相关性图不同，此处没有任何参数是经过估算的——该拓扑结构即为电网结构。DE LU 网络拥有 11 条互联线路，是整个网络中的天然枢纽。

0.20、8.25的OOS Sharpe比率、其综合值约为最佳单输入值的100倍，以及四个门中全部通过。

该同一套模型在股票市场中获得的夏普比率约为 1.4。如此优异的结果并非某种新发现，而是某种现象的体现。经诊断，发现存在三个复合性伪影。首先，标签重叠问题：将 24 小时远期收益率按小时采样后，连续观测值中有 23 小时是重叠的——测得的滞后 1 自相关系数为 0.91——而夏普比率计算却将这些观测值视为独立样本，从而导致分母被低估了一个数量级。其次，“信号”本身较为微弱：一个仅由当前现货价格负值构成的单变量预测因子，其横截面投资线索系数为 0.235，高于 GAT 模型的 0.20。无论该模型究竟学习到了什么，其表现均受确定性的日间均值回归因素主导。第三，年度化常数采用了股票市场的年度化常数（每年 252 个观测期）并应用于每小时采集的 8,760 个观测点。此外，还存在一个与市场结构相关的前瞻性因素：日前拍卖会会将 D 日的全部 24 小时数据一次性公布于 D_{-1} 日，因此这种“每小时”视角下的 24 小时前瞻标签在某种程度上属于同期信息。

该报告旨在强调的一个核心观点是：事件本身的规模本身便构成了警报信号。正是这种对合成对照组方法持怀疑态度的态度，使得该真实数据所呈现的数值被视为不可信，而非公布一个高达 8 的夏普比率。

6.3 日前调度框架下的重新设计

此次重新设计将评估时间点调整至市场实际的决策频率。日前竞价会每日进行一次，因此将当前时点设为交割日：即基于截至 D 日的数据所计算的基荷聚合特征；该标签等于真实未知的次日基荷价格变化量 $D \rightarrow D+1$ ，且根据其构建方式互不重叠，经年化处理（年化因子为 365）。

所有机械性检验结果均显示正常。Label Lag-1 自相关系数从 0.91 下降至 0.10；随机标签对照组得出的

the shuffle-label control gave a validation IC of 0.07 against 0.19 for real labels, so the remaining signal is not leakage. And yet the result stayed implausible: GAT composite OOS Sharpe 11.08 — with the trivial negative-price predictor at 6.15. The interim reading was that the pipeline was now honestly measuring a real but non-tradeable physical regularity: the law of one price across interconnected zones, already priced by cross-border capacity auctions. That reading was half right, and its other half is the most instructive result in this report.

6.4 The third artifact: a clip on the evaluation return

A direct challenge — *have you actually confirmed the strategy would lose nothing?* — prompted one more check: recompute the same strategies under different definitions of the evaluation return. Table 6.1 shows what that found.

Table 6.1: The return-definition diagnostic (real ENTSO-E daily frame, OOS). The ± 0.8 clip, inherited from the training label, was also applied to the evaluation return in E13 — and manufactured the entire positive result.

Evaluation return	trivial –price Sharpe	GAT Sharpe
floored + ± 0.8 clip (as trained)	6.280	11.000
winsorised ± 0.5	4.610	—
log return	3.750	—
plain relative return (honest)	–1.520	–1.540

The mechanism is specific and worth stating plainly. The long-short strategy shorts expensive zones. European power prices have fat right tails — scarcity spikes reaching 1,896 EUR/MWh in this sample — and the ± 0.8 clip on the evaluation return caps exactly those short-leg losses. Removing the clip turns a Sharpe-11 winner into a strategy that loses money, for the GAT and the one-line baseline alike. The rank-IC, meanwhile, stays at about 0.21 under both return definitions: cheap zones really do tend to rise relative to expensive ones, but ranking correctly is blind to tail magnitudes, and the tails are where this strategy’s money is lost. A positive IC coexisting with negative PnL is the transferable lesson — never transform the evaluation return, and never read rank quality as profitability. The equity track is unaffected: its evaluation return was never clipped.

The final verdict on the energy alpha question is therefore airtight and negative. Under honest returns the day-ahead cross-sectional strategy loses money (Sharpe about -1.5); every positive energy number along the way was an artifact — overlap, annualisation, then the return clip — and the GAT adds no tradeable value over a one-line baseline. The fix (evaluation on unclipped returns) shipped as the default, so the repository can no longer reproduce the inflated numbers.

6.5 The reframe: does the graph improve forecasts?

What survives E13b is a genuine question the data can answer: not “is there alpha?” but “does the interconnector graph improve forecast skill?” — measured with the ladder of Section 4.7 on six months of real ENTSO-E data (20 zones, 87,360 rows, with day-ahead load, wind and solar forecasts, realised load, and generation mix as drivers). Table 6.2 shows the node-level ladder, and Figure 6.2 draws it with one added ingredient per rung.

Three findings, in decreasing order of robustness. First, the graph carries real forecast value: uniform neighbour averaging over the interconnector topology lifts skill by $+0.131$ over the no-graph ridge with the same drivers. The synthetic negative control (independent zones) shows no such lift, so the gain is genuine cross-zonal coupling. This rung is pure numpy — no learned model is involved — which makes it the hardest number in the chapter to argue with.

验证 IC 为 0.07，而真实标签组的验证 IC 为 0.19，这表明剩余信号并非泄漏信号。然而，该结果仍显得不太合理：GAT 复合模型的 OOS Sharpe 指数为 11.08——而采用简单的负价格预测因子时该指数为 6.15。这一阶段性结论表明，该分析流程如今正准确地测出了一个真实存在但不可交易的物理性规律：即跨互联区域的“一价定律”，该规律已通过跨境运力拍卖机制得到定价。这一结论中有一半是正确的，而另一半则是本报告中最具启发性的发现。

6.4 第三个人工制品：评估回执上的夹子

一项直接质疑——*您是否真正确认该策略不会造成任何损失？*——促使我们进行又一次验证：在不同的评估回报定义下重新计算相同的策略。表6.1展示了此次验证的结果。

表 6.1：返回值定义诊断（实际 ENTSO -E 每日帧，OOS）。源自训练标签的± 0.8 剪切阈值亦应用于 E13 的评估返回值——从而生成了完整的正向结果。

评估返回值	平凡价格夏普比率	GAT Sharpe
地板 +± 0.8 个剪辑片段（基于训练数据）	6.280	11.000
winsorised ±0.5	4.610	—
对数收益率	3.750	—
普通相对收益率（诚实法）	-1.520	-1.540

该机制具有特殊性，值得明确阐述。该长空策略对高价区进行做空操作。欧洲电力价格具有明显的右尾特征——在本样本中，稀缺性峰值高达 1,896 欧元/兆瓦时——而评估收益率上± 0.8 的截断值恰好限定了这些做空头寸的损失范围。若移除该截断值，原本在 Sharpe-11 指标上表现优异的策略将变为亏损策略，无论对 GAT 还是单线基准而言皆然。与此同时，无论采用何种收益率定义，rank-IC 值均维持在约 0.21：低价区确实往往相对高价区呈现上涨趋势，但正确的排序方法对尾部波动幅度视而不见，而策略亏损恰恰源于这些尾部波动。正向 IC 与负向 PnL 共存这一现象提供了可迁移的经验教训——切勿对评估收益率进行调整，亦勿将排序质量等同于盈利能力。股票投资策略则不受影响：其评估收益率从未被截断。

因此，关于“能源α”问题的最终结论是确凿且负面的。在采用真实收益率的情况下，日-ahead横截面策略会产生亏损（夏普比率约为-1.5）；此前出现的所有正值能源指标均属人为假象——源于数据重叠、年度化处理以及收益率截取等因素——而GAT策略相较于单线基准并未带来任何可交易的价值。默认采用的修正方法（基于未截取收益率进行评估）现已启用，因此该数据集已无法再重现此前被夸大的数值。

6.5 重新构建框架：该图表能否改善预测效果？

E13b所揭示的问题是数据能够解答的一个真实问题：并非“是否存在α？”而是“互联网络图是否能提升预测精度？”——该问题通过第4.7节所述的梯度法，基于六个月的实际 ENTSO -E数据（涵盖20个区域、87,360行数据，以日前负荷、风能与太阳能预测值、实际负荷及发电结构作为驱动变量）进行衡量。表6.2展示了节点级梯度图，而图6.2则在每个梯级上额外添加了一个变量来绘制该图。

三项研究结果按稳健性由高到低排序如下：首先，该图具有真实的预测价值——在相同的驱动因素条件下，对互联线拓扑结构进行均匀邻域平均处理，可使预测性能较无图基准情况提升 +0.131。而合成负向对照组（独立区域）并未出现此类提升，因此该增益源于真实的跨区域耦合效应。这一结果完全基于纯数值计算（未涉及任何机器学习模型），因而成为本章中最难被质疑的数据。

Table 6.2: Node-level price forecast skill ladder ($\text{skill} = 1 - \text{MSE}/\text{MSE}_{\text{persistence}}$, OOS, $k = 24\text{h}$).

predictor	skill	rank ic	note
persistence	0.000	0.637	reference (tomorrow = today)
seasonal_naive	0.000	0.637	diurnal carry
no_graph_ridge	0.224	0.636	own physical drivers (ridge)
uniform_graph_ridge	0.355	0.584	+ interconnector-neighbour mean (unlearned graph)
gat_node	0.347	0.612	learned attention (PyG GATv2; 5-seed mean)
gat_congestion	0.346	0.615	+ price-spread congestion edge feature (5-seed mean)

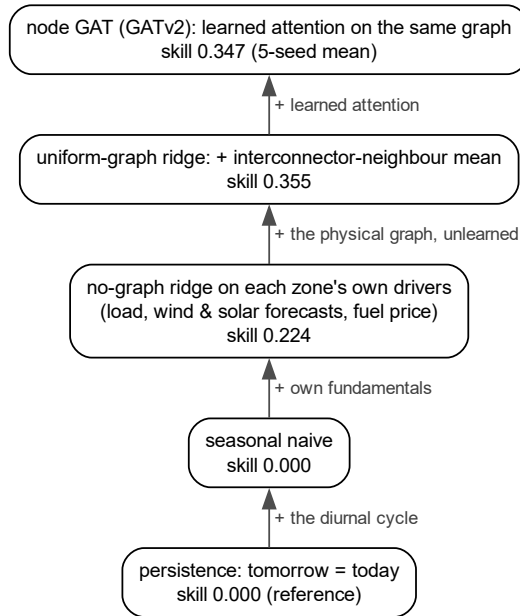


Figure 6.2: The node-level forecast ladder of Table 6.2, drawn as a ladder. Each rung adds exactly one ingredient, so each skill increment isolates one cause. The published drivers enter at the ridge rung and are shared by every rung above it; the graph’s contribution (+0.131) enters below any learning.

表 6.2: 节点级价格预测技能梯度 (技能 = $1 - \text{MSE}/\text{MSE}_{\text{persistence}}$, 外部数据集, $k=24$ 小时)。

预言者	技巧	排名 IC	笔记
持久性	0.000	0.637	参考 (明日 = 今日)
季节性朴素模型	0.000	0.637	日间运输
无图脊线	0.224	0.636	自定义物理驱动程序 (脊线)
均匀图脊线	0.355	0.584	+ 互联节点-邻节点均值 (未学习图)
GAT节点	0.347	0.612	学习注意力机制 (PyG GATv2; 5次随机种子均值)
通道拥塞	0.346	0.615	+ 价格价差拥堵边缘特征 (5个种子均值)

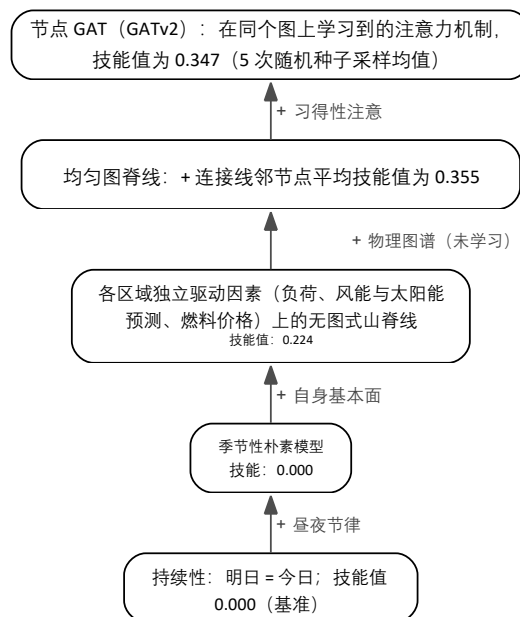


图 6.2: 表6.2中的节点级预测梯度图 (以梯子形式绘制)。每个梯级仅添加一个要素, 因此每个技能增量都对应着一个独立的驱动因素。已公布的驱动因子从最上层的梯级开始输入, 并被其上方的所有梯级共享; 而该图的贡献值 (+0.131) 则作用于任何学习行为之前。

Second, learned attention beats the uniform anchor on cross-sectional *ranking* in five seeds of five (rank-IC 0.61 against 0.58) but is a wash on MSE skill (better in two seeds of five). Uniform neighbour-averaging turns out to be near MSE-optimal on this graph — a hard bar — so the GAT’s robust edge here is order, not magnitude, echoing the equity finding that attention’s value is a modest tilt on broad pooling.

Third, the congestion hypothesis — that attention should down-weight saturated borders if told how congested they are — is not supported. Two operationalisations of the edge feature were tried: the current cross-border price spread (a leak-safe proxy), and ground-truth physical flows over day-ahead transfer capacity fetched for all 38 borders. The proxy’s apparent gain existed only in a hand-rolled dense GAT implementation (+0.031 skill, five of five seeds) and vanished under the standard PyG GATv2 (−0.002, two of five); the ground-truth version beat node-only attention in one seed of five. The hypothesis is not refuted — capacity data covered only 36% of borders, and the most coupled region uses flow-based coupling that a flow-over-capacity ratio cannot express — but as a smooth additive edge feature, congestion does not add forecast skill. The two-backend check deserves its methodology note: a single-seed PyG run had initially shown +0.072, a lucky draw that a five-seed rerun corrected, and the cross-implementation comparison is what stopped a fragile effect from becoming a claim.

6.6 Edge-level spread prediction: the relational payoff

The last experiment moved the target instead of the model. A cross-border price spread $\text{spot}_a - \text{spot}_b$ is irreducibly relational — it does not exist for a single node, and it is the object that transmission rights actually price. A GAT computes node embeddings by message passing over the interconnector graph; a small MLP head predicts each border’s next-day spread from the two endpoint embeddings and the current spread. The baseline that matters is a ridge that already sees both endpoints’ drivers and the current spread — so any gain over it is whole-network context, not information the baseline lacks. Table 6.3 shows the ladder over the 38 borders.

Table 6.3: Edge-level cross-border spread prediction (38 borders, OOS, $k = 24\text{h}$, 5-seed means).

predictor	skill	rank ic	note
edge_persistence	0.000		spread today (reference)
edge_ridge	0.192	0.67	both endpoints’ day-ahead drivers + current spread (no graph)
edge_gat	0.248	0.68	GAT node embeddings → edge MLP head (message passing; 5-seed mean)

The edge GAT beats the both-endpoint ridge by +0.056 skill — about 29% relative — in five seeds of five, and the synthetic control inverts as it should: on independent zones the same architecture is *worse* than the ridge, so the real-data gain is structure, not capacity. This is the project’s strongest relational result, and its placement is the point: node-level price forecasting barely needed the graph, congestion-as-edge-feature was null, and the GNN’s value concentrates precisely on the target that cannot be defined without the relation. Caveats stated once: a single six-month window, untuned hyperparameters, and modest absolute skill — spreads are noisy differences.

6.7 Summary of the energy evidence

Table 6.4 condenses the track. As an alpha project the energy track concluded negatively three artifacts deep, and the honest number — the strategy loses money — is stated as the finding it is. As a forecasting project it produced two robust, control-validated positives (the graph lift of

其次，在五组种子数据的横截面 *排序* 任务中，学习注意力法的表现优于均匀锚定法（rank-IC 值分别为 0.61 对比 0.58）；但在 MSE 技能任务中两者表现相当（在五组种子数据中有两组中学习注意力法表现更优）。在该图结构（一项具有挑战性的基准任务）中，均匀邻域平均法被证明近乎 MSE -最优；因此，GAT 模型在此方面的突出优势在于“顺序性”而非“数值大小”，这与关于“注意力机制的价值是对广义池化方法进行适度倾斜”的公平性发现相呼应。

第三，拥堵假说——即当告知某个饱和边界所处的拥堵程度时，注意力应降低对该边界的权重——并未得到支持。研究尝试对边缘特征进行了两种操作化处理：当前跨境价格价差（一种防泄漏的代理变量），以及基于日前传输容量为全部38个边界获取的真实物理流量数据。该代理变量所呈现的明显增益仅出现在手动实现的密集型GAT算法中（+0.031技能值，5次种子均有效），而在标准PyG GATv2算法中该增益消失（-0.002,5次种子中仅有2次有效）；而真实数据版本在5次种子中仅有1次优于纯节点注意力机制。这一假说并未被推翻——因为容量数据仅覆盖了36%的边界，且耦合最紧密的区域采用了基于流量的耦合方式，而这种耦合无法通过流量与容量之比来表征——但作为一种平滑的加性边缘特征，拥堵因素并未提升预测技能。对两种后端实现的对比分析值得特别说明其方法论依据：单次PyG算法运行最初显示+0.072的增益，这一偶然结果经五次种子的重复运行后得到修正；而通过跨实现方案的对比，才避免了将这一脆弱效应误作定论。

6.6 边缘层面的扩散预测：关系性收益

最后一项实验中，移动的是目标对象而非模型。跨境价格利差现货 _

Spot B 是不可约关系型变量——它并非针对单个节点存在，而是传输权值实际定价的对象。GAT 通过在互联图上进行消息传递来计算节点嵌入表示；一个小型 MLP 头部根据两个端点的嵌入表示及当前价差来预测每个边界节点的次日价差。关键的基准模型是一个能够同时捕捉两个端点驱动因素及当前价差的岭回归模型——因此，相较于该基准模型所获得的任何增益均源于全网络上上下文信息，而非基准模型所缺失的信息。表6.3展示了覆盖 38 个边界的梯度分布情况。

表 6.3：边缘级跨境蔓延预测（38 个边界，OOS，k= 24 小时，5 个种子均值）。

预言者	技巧	排名 IC	笔记
边缘持久性 (Edge Persistence)	0.000		今日价差 (参考)
——边缘脊线	0.192	0.67	两个端点的日前驱动因子 + 当前价差 (无图示)
边缘GAT	0.248	0.68	GAT节点嵌入 → 边 MLP 头部 (消息传递; 5种子均值)

在五组种子数据中，Edge GAT 比双端点 Ridge 模型高出 +0.056 个技能分（约相当于 29% 的相对优势）；而合成对照组则如预期般表现逊色：在独立数据集上，该相同架构的表现 *比 Ridge 模型更差*，因此这种性能提升源于数据结构而非模型容量。这是本项目中最具影响力的关联性研究成果，其排名恰恰揭示了核心要点：在节点级价格预测任务中，图结构几乎并非必需，将网络拥塞作为边特征的做法效果为零，而 GNN 的价值 *precisely* 体现在那个若缺乏关联性便无法定义的目标上。需要特别说明的是：本研究仅采用单个六个月时间窗、未调优的超参数且绝对技能分值较低——所呈现的价差差异实则带有噪声。

6.7 能源证据摘要

表6.4汇总了该跟踪分析结果。作为一项 Alpha 项目，该能源跟踪分析最终得出结论：存在三个负面人工痕迹；而“真实数据”——即该策略导致亏损——则如实反映了这一研究结果。作为一项预测性项目，该分析产生了两项稳健且经控制验证的正向结果（即“图表提升”现象）。

Table 6.4: Energy forecasting track: what is and is not supported (real ENTSO-E data, 20 zones, 5-seed means).

finding	comparison	skill delta	seeds	verdict
Graph helps price level	uniform - no-graph	+0.131	control-validated	robust
Attention vs uniform (ranking)	gat_rank_ic - uniform	+0.031	5/5	robust
Attention vs uniform (skill)	gat - uniform	-0.008	2/5	wash
Congestion edge feature (price-spread proxy)	gat_congestion - gat_node	-0.002	2/5	null
Congestion edge feature (real flow/NTC)	flow - node	-0.005	1/5	null
Edge-level spread (message passing)	edge_gat - edge_ridge	+0.056	5/5	robust *

+0.131 at node level; the edge-level message-passing gain of +0.056, five of five seeds) and one clean null (congestion edge features, under both operationalisations). As methodology it produced the report’s most transferable lessons: evaluate on untransformed returns, distrust magnitudes that outrun their plausibility class, and require cross-implementation replication before believing a learned effect.

7 Discussion

Every limitation below is a deliberate scope choice or a measured finding, not an oversight discovered in retrospect; several are results in their own right. This chapter names each threat to validity in the standard vocabulary and states where the project mitigated it, and where it simply lives with it.

Look-ahead bias. The report contains a full worked example of this threat rather than merely a disclaimer: the energy track’s first real-data result (Section 6.2) was look-ahead bias in two of its three layers, and the diagnostic arc that dismantled it is the chapter. On the equity track one mild instance is disclosed and retained: the static correlation graph is built from data up to the split date, so early *training* snapshots see a graph estimated partly from their own future. The out-of-sample window is unaffected — the graph strictly predates it — and the clean alternative, the dynamic graph, was rejected because it performs worse (Section 5.3). Disclosure was chosen over a cosmetic fix that costs real out-of-sample performance.

Survivorship bias. The equity universe is roughly fifty hand-picked, currently listed US large caps fetched from a free vendor. Names that failed or delisted over 2021–2026 are absent, which inflates the absolute level of every Sharpe in Chapter 5. The core claim is deliberately insulated: the attention A/B compares models on the *same* biased universe, so the bias raises both sides of the comparison rather than manufacturing the gap. Absolute levels should be read as optimistic; relative margins are the load-bearing numbers. A survivorship-bias-free vendor is the single highest-value data upgrade a successor project could make.

Overfitting and data snooping. Ten alphas, one composite, and repeated evaluation on one out-of-sample window invite snooping; the four gates exist to police exactly this, and the hyperparameter search was designed so the out-of-sample window could not influence selection (Section 5.5). The residual risks are stated rather than waved away: the walk-forward advantage is significant only within the tuned configuration ($n = 5$ per arm) and is otherwise a directional claim; the flat

表 6.4：能源预测轨道：支持与不支持的内容（真实 ENTSO -E 数据，20 个区域，5 个种子均值）。

调查	比较	技能差值	种子	裁决
图表用于确定价格水平： 关注型 vs 均匀型（排序型）	均匀分布 - 无图 GAT Rank IC - 统一版	+0.131 +0.031	经对照验证 5/5	强健的 强健的
注意力 vs 统一（技能）	GAT - 制服	-0.008	2/5	洗
拥堵边缘特征（价格价差代理指标）；拥堵边缘特征（实际流量 / NTC）；边缘级价差（消息传递法）	GAT拥堵节点 流节点	-0.002 -0.005	2/5 1/5	零值的 零值的
	边缘门控 - 边缘脊线	+0.056	5/5	鲁棒 *

节点层面增益为 +0.131；边层面消息传递增益为 +0.056（五个种子节点全部适用）；以及一个干净的零值样本（基于拥塞边特征，在两种操作化方式下均适用）。从方法论角度来看，该研究得出了本报告中最具普适性的启示：应在未经转换的回报数据上进行评估；对超出其合理性范围的效应幅度持审慎态度；在认可某个学习到的效应之前，需进行跨实施环境的重复验证。

7 讨论

下文所列的各项限制条件均属有意为之的范围设定或经过审慎考量后得出的结论，而非事后发现的疏忽；其中部分限制条件本身即构成某种结果。本章采用标准术语对每一项效度威胁进行命名，并说明项目在哪些方面对这些威胁进行了缓解，而在哪些方面则直接承受了这些威胁。

前瞻偏差。本报告提供了这一威胁的完整实操案例，而非仅仅发布免责声明：在能源追踪路径中，首个基于真实数据得出的结果（第6.2节）在其三个层中的两个层中存在前瞻偏差；而用于揭示这一问题的诊断性分析段落即为本章内容。在股票追踪路径中，报告披露并保留了一个较轻微的实例：静态相关性图是基于截至拆分日的数据构建的，因此早期的训练快照所看到的图谱部分是基于其自身未来的数据估算得出的；样本外窗口不受影响——该图谱严格早于样本外窗口构建——而经过清理的替代方案（动态图谱）因表现较差而被拒绝（第5.3节）。本报告选择公开披露该问题，而非采用一种会损害实际样本外性能的表面性修正方案。

存活偏差。本研究的股票样本集大致由50只经人工筛选、目前在美国上市的大型股组成，这些数据源自一家免费供应商。在2021–2026年间遭遇失败或被退市的股票均未纳入样本集，这导致第5章中所有夏普比率的绝对值均被高估。本文的核心论点经过刻意保护：A/B组对比模型所采用的样本集相同且均存在偏差，因此该偏差实际上抬高了对比双方的数值，而非人为制造差距。这些绝对值应视为乐观估计；而相对边际值才是承载实际分析意义的关键数据。采用无生存偏差的供应商，是后续项目所能实施的单项最高价值的数据升级举措。

过拟合与数据窥探问题。十个 α 参数、一个复合参数以及在单个样本外窗口上进行的重复评估容易引发数据窥探现象；而四个门控机制正是为了精准防范此类问题设立的，且超参数搜索的设计使得样本外窗口无法影响参数选择（第5.5节）。残余风险被明确指出而非回避：步进预测优势仅在经过调参的配置下（每臂 $n=5$ ）才具有显著性，否则这仅是一个方向性结论；平坦的超参数曲

hyperparameter surface cuts both ways, as robustness and as a warning that tuning gains here are mostly noise; and the energy forecasting results rest on a single six-month window with untuned hyperparameters. The two synthetic negative controls and the shuffle-label control (Sections 5.1 and 6.1) are the strongest argument that the pipeline as a whole does not snoop.

Transaction costs and tradability. No result in this report is net of costs: there is no fill model, no spread, no market impact. The equity Sharpe of 1.37 is a paper Sharpe on daily bars, and a five-day-horizon long-short over fifty names would face real turnover costs. The energy track makes the point more brutally — its strategy loses money *before* any costs are applied (Section 6.4), and the E13b lesson (rank quality does not imply profitability when tails carry the money) would only sharpen under a cost model. The platform’s stated production boundary (Section 3.1) is consistent with this: nothing here is a trading system.

Regime dependence and external validity. All equity claims rest on a single 5.5-year window of one market regime, at one horizon ($k = 5$ days), on one universe; all energy forecasting claims on six months of one calendar year, at one horizon ($k = 24$ hours), on twenty zones. The attention analysis itself argues against over-generalising: the learned structure is temporally stationary (Section 5.6), so the project’s own evidence says the model found a stable statistical regularity, not a regime-adaptive mechanism. Whether either result survives a different half-decade is untested.

Reproducibility. What is reproducible is the distribution, not the point: the same seed produces different results on different devices, for identified mechanical reasons (Section 5.4). All headline numbers are therefore means over at least five seeds on a fixed device, the test suite pins the leakage-critical logic, and the archived per-run artifacts allow every table in this report to be regenerated. Bit-exact cross-device replication is not claimed and, on CUDA, not attainable.

The open gate. The relational composite never beat the best single island alpha — an OOS Sharpe of roughly 1.4 against roughly 3 — and this report resists the temptation to soften that bar post hoc. Two softer readings are available and honestly labelled as softer: the composite beats nine of its ten inputs, and it is nearly uncorrelated with all of them (maximum 0.26), which for a practitioner suggests portfolio value that the max-of-singles gate does not measure. A marginal-contribution-to-a-multifactor-portfolio reading is the natural next evaluation, planned but not executed within this project’s scope.

8 Conclusion and Future Work

The three research questions of Chapter 1 can now be answered directly.

Does learned attention add value over unlearned relational propagation? On the equity track, yes, with unusual consistency for a result in this domain: positive attention value-add in thirty of thirty seeded runs, across two hyperparameter configurations and two devices, against an anchor that shares the model’s inputs and topology. The mechanism is modest and now understood — a stationary, near-uniform reweighting of neighbour pooling — and the resulting composite is a real, unique, consistent factor that nevertheless does not beat the best single alpha it was built from. On the energy forecasting track the answer is narrower: attention improves cross-sectional ranking over the uniform anchor in five of five seeds but not mean-squared skill, where uniform pooling is already near-optimal.

Does one kernel transfer across heterogeneous graphs without manufacturing false positives? Yes, and the negative controls are the evidence: the same code that finds a consistent positive margin on real equity data finds nothing on synthetic random walks, finds nothing on synthetic uncoupled power zones, and — when real energy data produced numbers too good to be true —

面具有双面性——既体现了模型的鲁棒性，也警示着此处通过调参获得的收益大多属于噪声；而能源预测结果基于一个采用未调参超参数的单一六个月时间窗口。两个合成阴性对照组及随机标签对照组（第5.1节与第6.1节）构成了最有力的证据，表明整个分析流程并未存在数据窥探现象。

交易成本与交易流动性。本报告中的所有结果均未扣除交易成本：不存在执行模型、不存在价差、亦不存在市场冲击效应。股票的 Sharpe 指数为 1.37，这是基于每日数据条计算的理论 Sharpe 指数；而对 50 只股票进行为期五天的多空策略投资，则会面临实际的交易成本。能源板块的案例则更直观地揭示了这一点——其策略在计入任何成本之前就已出现亏损（第6.4节），而 E13b 所揭示的经验教训（当极端行情主导资金流向时，排名质量并不等同于盈利能力）在引入成本模型后将更为凸显。该平台所声明的生产边界（第3.1节）与这一结论一致：此处没有任何内容属于交易系统。

经济制度依赖性与外部有效性。所有权益类预测均基于单一、为期 5.5 年、处于单一市场制度、在单一时间跨度（ $k=5$ 天）且针对单一样本集的分析窗口；而所有能源类预测则基于一个日历年内的 6 个月数据、在单一时间跨度（ $k=24$ 小时）且针对 20 个区域的分析。该注意力机制分析本身旨在反对过度泛化：所构建的结构在时间上是平稳的（第5.6节），因此该项目自身的证据表明，该模型发现的是一个稳定的统计规律性，而非一种适应经济制度变化的机制。上述任一结论在另一段为期五年的时间跨度内是否依然成立，尚待检验。

可复现性。真正可复现的是数据分布，而非单个数据点：由于已明确的机械性原因（第5.4节），相同的种子在不同设备上会产生不同的结果。因此，所有主要数值均为在固定设备上对至少五个不同种子计算所得的均值；测试套件针对泄漏敏感逻辑进行了精准定位；而存档的每次运行生成的工件文件使得本报告中的每一张表格均可重新生成。本报告并未声称实现位精确的跨设备复制，且在 CUDA 平台上，这种复制并不可行。

开放式门限。该关系合成指标从未跑赢最佳单一岛屿 Alpha 指标——其 OOS Sharpe 比率约为 1.4，而最佳单一指标约为 3——本报告并未受诱惑对这一基准进行事后调低。目前提供两种较弱的解读结果，且均诚实地标注为“较弱”：该合成指标在十项输入指标中跑赢了九项，且与所有输入指标几乎不相关（最大相关系数为 0.26）；对于从业者而言，这表明了“单一指标最大值门限”所无法衡量的投资组合价值。采用 marginal-contribution-to-a-multifactor-portfolio 指标进行评估是顺理成章的下一步，该评估虽已规划，但未纳入本项目的实施范围。

8 结论与未来工作

第1章中的三个研究问题现在可以直接得到解答。

*学习到的注意力机制是否比非学习到的关系传播机制更具价值？*在公平性评估赛道上，答案是肯定的——这一结果在该领域中实属罕见：在两种超参数配置和两种设备上进行的 30 次随机种子实验中，针对一个与模型输入和拓扑结构相同的基准模型，注意力机制均带来了正向的价值增益。其作用机制较为温和且现已得到阐明——即对邻域池化进行一种静态、近乎均匀的重新加权操作——而由此产生的复合因子是一个真实、独特且具有一致性的因子，但其表现并未超越其构建所依据的最佳单参数 α 值。在能源预测赛道上，答案则较为局限：在 5 次随机种子实验中，注意力机制在横截面排名上优于均匀池化基准模型，但在均方误差指标上则未见优势，因为在该场景下均匀池化已接近最优。

*是否可以在不产生假阳性结果的情况下，将单个模型参数迁移至异构图上？*答案是肯定的，而阴性对照便提供了证据：同一个在真实股权数据上能找到一致正边际的代码，在合成随机游走数据上未能发现任何规律，在合成非耦合功率区域数据上也未能发现任何规律；而当真实能源数据得出的结果好得令人难以置信时，相关

the surrounding discipline treated the magnitude as an alarm rather than a result. The transfer claim is literal: one model class, one training loop, one gate suite, two structurally different graphs.

Does the physical topology improve forecasts? Yes, where it should. The interconnector graph lifts node-level price forecast skill by +0.131 over an identically-informed no-graph baseline, and the clearest relational gain sits on the irreducibly relational target: edge-level spread prediction beats a both-endpoint baseline by +0.056 skill in five of five seeds, while a synthetic control inverts the comparison. Congestion as an explicit edge feature added nothing robust under either of two operationalisations — an honest null.

Beyond the questions it set out to answer, the project’s most transferable output is methodological: evaluate on untransformed returns; treat rank correlation and profitability as different claims; treat implausible magnitudes as alarms; require seeds, fixed devices, and cross-implementation replication before believing a learned effect; and keep negative controls running permanently, because they are what give positive results their meaning.

Future work

Four directions follow directly from the open ends documented in the experiment log and forecasting notes, in decreasing order of expected value. First, data: a survivorship-bias-free equity vendor would turn the biased absolute Sharpe levels into defensible ones, and longer, multi-window energy samples would test the six-month forecasting results out-of-sample in time. Second, evaluation: the value-added gate deserves its portfolio-level variant — the marginal contribution of the composite to a multifactor portfolio — alongside the max-of-singles bar it never cleared. Third, the forecasting track is untuned: a hyperparameter grid, seed ensembles, and walk-forward retraining, all already built for the alpha track, transfer mechanically; an edge-level congestion target (not just the spread) is the natural next prediction problem. Fourth, the congestion hypothesis remains open rather than dead: the flow-based-coupling regions publish shadow prices on critical network elements, and a congestion signal built from those — rather than from flow-over-capacity ratios with partial coverage — is the experiment this project would run next. Any renewed energy *alpha* attempt, by contrast, should wait for genuinely tradeable targets: balancing or intraday spreads, or transmission-rights-aware profit accounting.

The report closes where it began. The coupling structure of markets is information; on the evidence here, learning how to read it earns a consistent but modest premium over reading it naively, and the discipline required to measure that premium honestly — anchors, controls, embargoes, seeds, and the willingness to let three spectacular numbers die — is itself the result this project most confidently recommends to its successors.

领域学者将该数值视为一种警报而非结论。这一“迁移”主张是字面意义上的：一个模型类、一个训练循环、一套门控机制，以及两个结构不同的图。

物理拓扑结构是否能提升预测精度？是的——正如其理。在信息对称的无拓扑结构基准模型基础上，互联线路图将节点层面的价格预测能力提升了 +0.131；而最显著的关系型增益则体现在不可约的关系型目标上：在全部五组种子数据中，边层面的价差预测能力均比两端点基准模型高出 +0.056；而合成对照组则使这一对比关系发生逆转。将网络拥塞作为显式边特征进行引入，在两种不同的操作化方式下均未带来任何稳健的增益——这相当于一个纯粹的零假设。

除了该项目旨在解答的具体问题外，其最具普适性的产出在于方法论层面：在未经处理的收益率上进行评估；将秩相关性与盈利能力视为不同的研究命题；将不合理的数值视为警示信号；在确认某个学习到的效应之前，必须进行种子设置、固定装置设置及跨实施环境的重复验证；并且应持续运行阴性对照组——因为正是这些阴性对照组赋予了阳性结果以意义。

未来工作

以下四个方向直接源自实验日志与预测笔记中记录的开放端点，并按预期值降序排列：首先，数据层面：一家无生存偏倚的股票投资机构可将存在偏倚的绝对夏普比率转化为可辩护的比率；而更长、多窗口的能源样本则可在时序上对六个月期的预测结果进行样本外检验。其次，评估层面：增值门应当配备其组合层面的变体——即该综合指标对多因子投资组合的边际贡献——与此前从未超越的单因子最大值指标并列。第三，预测轨道尚未调校：超参数网格搜索、种子集成法及前向走步式重新训练——这些方法均专为阿尔法策略轨道设计——可机械迁移应用；而边缘层面的拥塞目标（而非单纯的价差）是自然的下一个预测课题。第四，拥塞假说仍具研究价值而非已被否定：基于流量耦合的区域可公布关键网络节点的影子价格；而基于这些影子价格构建的拥塞信号——而非基于覆盖范围不完整的流量超载比率——将是本项目接下来要开展的实验。相比之下，任何新的 α 投资尝试都应等待出现真正可交易的标的：例如平衡价差或日内价差，或基于输电权的利润核算。

本报告以开篇之笔作结。市场间的耦合结构本质上就是信息；根据本文所呈现的证据，学会解读这种信息，相较于简单地照单全收，能够带来稳定但有限的超额收益；而要诚实地衡量这一超额收益所需的严谨态度——包括运用锚定效应、控制变量、信息封锁、种子数据，以及愿意让三个看似亮眼的数字失效——正是本项目最 confidently 建议给后继者们践行的准则。

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A Reproducibility

Everything claimed in this report can be regenerated from the repository.¹ This appendix records the environment, the commands, and the mapping from experiment to archived artifact.

Environment

Python 3.13 with `torch 2.12.0` and `torch_geometric` (installed via the project’s `[gnn]` extra); `PYTHONPATH` set to `src`. All headline experiments ran on CPU — a deliberate choice after the device study of Section 5.4 — with the device recorded per experiment. The equity track fetches market data through `yfinance`; the energy track runs on the synthetic generator by default and on live ENTSO-E data when an API token is supplied via the `ENTSOE_API_KEY` environment variable. The warehouse and marts require `duckdb`, `dbt-core`, and `dbt-duckdb`; the GAT pipeline itself does not.

Tests

The full suite is 115 tests; 55 of them cover the capstone modules (graph construction, training primitives, propagators, factor providers, both track runners, attention analysis, and the leakage controls). The leakage controls — shuffle-label negative control and planted-signal positive control — are pinned to CPU for determinism and run on every change. `py -m pytest` from the repository root executes everything.

Experiments and artifacts

Each experiment entry in the log (`docs/gat_experiment_log.md`, entries E1–E14) names its run script and its archived outputs. The run scripts are one-off drivers kept in an untracked scratch area outside the published repository; each log entry records the configuration they ran — model variant, seeds, hyperparameters, device — and the CLI commands below drive the same pipelines. The outputs — per-run diagnostics, matrix, seed, hyperparameter-grid, winner-validation, energy, and attention CSV/JSON files, plus five attention figures — are date-prefixed under `docs/results/`. The canonical trained equity weights are tracked at `data/warehouse/gat_equity.pt`. Every quantitative table in this report is generated mechanically from those archived CSVs; no result number in the text was typed by hand from memory.

Command reference

The equity pipeline:

```
quant-alpha gat-equity --graph {static,dynamic}
                        --retrain {single,walk-forward}
                        --loss {ic,mse}
                        --device {cpu,cuda,auto}
                        --persist
```

¹<https://github.com/witold-andelie/quant-foundation-gat>

可重复性

本报告中所述的所有内容均可从存储库中重新生成。¹本附录记录了实验环境、执行命令以及实验与归档制品之间的映射关系。

环境

使用 Python 3.13、torch 2.12.0 及 torch_geometric（通过项目中的 [gnn] 额外包安装）；Python 路径设置为 src。所有核心实验均在 CPU 上运行——这是在第 5.4 节设备研究之后做出的有意选择——每个实验均记录了设备信息。Equity 领域通过 yfinance 获取市场数据；Energy 领域默认使用合成发电机进行运算，当通过 ENTSOE_API_KEY 环境变量提供 API 令牌时，则使用实时 ENTSOE 数据。Warehouse 和 Mart 部分需要安装 duckdb、dbt-core 和 dbt-duckdb；而 GAT 管道本身无需安装这些依赖项。

试验

该完整测试套件共包含 115 项测试；其中 55 项涵盖顶点模块（图构建、训练原语、传播器、因子提供器、两种轨迹运行器、注意力分析以及泄漏控制）。泄漏控制项包括：随机标签负向控制项与植入信号正向控制项。——这些测试被固定到 CPU 上以确保确定性，并在每次代码变更时运行。从代码库根目录执行 `py -m pytest` 命令即可运行所有测试。

实验与人工制品

日志中的每个实验条目（docs/gat_experiment_log.md，条目 E1–E14）均标注了对应的运行脚本及其归档输出结果。这些运行脚本为单次执行的驱动脚本，存储在发布仓库之外的、未被纳入版本控制的临时存储区中；每个日志条目均记录了所执行的配置——包括模型变体、随机种子、超参数及设备——而下方的 CLI 命令则用于驱动相同的实验流程。输出结果——包括每次运行的诊断信息、矩阵、随机种子、超参数网格、胜者验证结果、能耗及注意力机制相关 CSV/JSON 文件，以及五幅注意力机制示意图——均按日期命名并存放于 docs/results/ 目录下。标准训练后的股票权重数据记录于 data/warehouse/gat_equity.pt。本报告中的所有定量分析表格均通过自动脚本从上述归档的 CSV 文件中生成；文中所有结果数值均非凭记忆手动输入。

命令参考手册

股权融资渠道：

```
Quant-Alpha GAT-Equity --graph {静态, 动态}
                        --重新训练 {单次、前向走步}
                        -损失 {ic, mse}
                        -设备 {CPU、CUDA、自动}
                        --持久化
```

¹<https://github.com/witold-andelie/quant-foundation-gat>

runs end to end and, with `-persist`, writes the four warehouse tables of Section 3.3. The energy alpha runner is `run_gat_energy` (synthetic or `source="entsoe"`); the forecasting ladder is `quant-alpha energy-forecast -source {synthetic,entsoe} -k 24` with a `-raw-path` Parquet cache for repeatable real-data runs. The marts rebuild with `dbt build -profiles-dir .` inside `dbt_quant_alpha/` (21 models and tests) and `dbt_energy_alpha/` (12). The report's build script regenerates the tables and figures from the repository's artifacts and reference data and then runs `pdflatex` and `biber`.

What reproducibility means here

As established in Section 5.4, bit-exact replication across devices is not attainable for CUDA-adjacent training stacks and is not claimed. Rerunning any experiment with the recorded seeds on a CPU of the same class reproduces the reported distributions — means and standard deviations over five or more seeds — and every individual archived run is identified by seed, device, configuration, and date in its artifact file.

该流程实现端到端运行；当使用 `-persist` 参数时，该流程将写入第3.3节中所述的四个仓库表。能源 Alpha 运行程序为 `run_gat_energy`（合成数据或 `source= "entsoe"`）；预测层级结构为 `quant-alpha energy-forecast`（参数：`-source {synthetic, entsoe} -k 24`），并配置了 `-raw-path` Parquet 缓存，以便于进行可重复的真实数据运行。数据集的重建操作通过 `dbt build --profiles-dir` 命令执行。

· 位于 `dbt_quant_alpha/`（包含 21 个模型与测试）和 `dbt_energy_alpha/`（包含 12 个模型）目录下。该报告的构建脚本会根据仓库中的构建产物及参考数据重新生成相关表格与图表，随后执行 `pdflatex` 和 `biber` 命令。

此处“可重复性”指的是什么？

如第5.4节所述，对于 CUDA 相邻训练堆栈而言，实现跨设备的位精确复制并非可行，也未对此作任何承诺。若在同类别 CPU 上使用记录的种子重新运行任何实验，均可复现所报告的分布（即基于五个或更多种子计算得出的均值与标准差）；且每个单独存档的运行结果均在其 artifact 文件中通过种子、设备、配置及日期进行标识。